

A toolbox for genetic targeting of the claustrum

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Abstract

The claustrum (CLA), a subcortical nucleus in mammals, essentially composed of excitatory projection neurons and known for its extensive connections with the neocortex, has recently been associated with a variety of functions ranging from consciousness to impulse control. However, research on the CLA has been challenging due to difficulties in specifically and comprehensively targeting its neuronal populations. In various cases, this limitation has led to inconsistent findings and a lack of reliable data. In the present work, we describe the expression profile of the *Smim32* gene, which is almost exclusively transcribed in excitatory neurons of the CLA and the endopiriform nucleus, as well as in inhibitory neurons of the thalamic reticular nucleus. Leveraging this unique expression pattern, we developed a series of Cre- and Flippase-expressing knockin and BAC transgenic mouse lines with different expression profiles. With these novel tools in hand, we propose new standards for the interrogation of CLA function.

eLife assessment

This **valuable** research identifies *Smim32* as a new genetic marker for the claustrum and generates transgenic mouse lines aimed at enhancing specificity when studying this brain region. However, the evidence supporting the increased specificity of this marker and its associated transgenic lines is **inadequate**, as *Smim32*'s specificity to the claustrum is limited. Nevertheless, this work will be of interest to researchers studying the molecular organization of the claustrum.

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Introduction

The CLA is a small, sheet-like, subcortical structure nestled between the insular cortex and the striatum in mammals. Known for its extensive connectivity, it has been described as having the densest reciprocal connections with the neocortex¹⁻³, and exhibits anteroposterior and dorsoventral patterning of these connections with different neocortical areas⁴⁻⁸. Given its

unique connectivity map, the CLA has recently sparked broad interest, and been attributed various functions, including a role in consciousness^{9,10}, saliency detection^{11,12}, and synchronization of neuronal activity¹³. Recent functional studies, primarily in mice, have suggested roles in a wide range of activities and processes such as behavioral flexibility^{14,15}, attention^{16,17}, sleep^{18–20}, contextual fear conditioning^{21–23}, top-down action control²⁴, impulse control^{25–27}, modulation of cortical activity²⁸, pain processing^{29–31}, contextual association of reward³², and behavioral stress response³³, among others. While the multifunctional role of the CLA is increasingly recognized, its study has been challenged by difficulties in accurately targeting the structure in murine models due to its complex topology. Unlike the human CLA, which is demarcated by two bundles of white matter fibers, making it clearly distinguishable, the mouse CLA boundaries are unclear: it extends half-way through the brain along the anteroposterior axis with varying spread and cell density¹, and its demarcation with the insular cortex laterally and corpus callosum medially is not well defined.

To explore CLA function, both historical and contemporary studies have often resorted to direct injections of viruses or tracers into the CLA area. However, these approaches lack cell-type specificity and are, therefore, challenging to replicate.

The use of genetic tools enables precise labeling and manipulation of defined cell populations. This precision is achieved by using specific promoters to drive the expression of transgenes in the desired target cells. To this end, the development of recombinase-expressing mice with specificity in time and space, combined with the use of recombinase-dependent reporter tools, such as viral vectors or transgenic mice, has enabled a wide range of applications. The CLA is primarily composed of projection neurons. The latter express a combination of genes, each showing different degrees of specificity and coverage, such as *Nr4a2*, *Lxn*, *Car3*, *Oprk1*, *Slc17a6* and *Gnb4*^{1,15,34,35}. Most of these CLA markers are also expressed in the endopiriform nucleus (EP), ventral to the CLA, and in a subset of cortical layer 6 neurons, rostral and dorsal to the CLA.

A few Cre-driver mouse lines have been used to study the CLA, including *Egr2-Cre*¹⁷, *Gnb4-IRES2-CreERT2*³⁶, *Slc17a6-IRES-Cre*¹⁵, and *Tg(Tbx21-Cre)*¹⁸. Unfortunately, each of these transgenic lines suffers from at least one of the following weaknesses: 1) Cre expression is observed in a poorly defined cell population (in terms of localization or identity) in the CLA area; 2) Cre expression is limited to a subset of CLA projecting neurons (limiting the investigation to this subset); 3) Cre expression requires injection of hormones (potentially limiting behavioral experiments); 4) Cre is expressed in non-CLA populations in the brain in close vicinity to the CLA (specificity is thus difficult to achieve even when using locally-injected viral reporters); and 5) Cre is broadly expressed in the brain and only stops being produced in the area surrounding the CLA during postnatal development (constraining its combination with viral approaches and precluding comprehensive CLA targeting). Overall, establishing a standardized, specific, and exhaustive targeting of CLA cells for its functional interrogation remains challenging today.

Our goal was here to generate genetic tools capable of targeting the majority of mouse CLA projection neurons without affecting other brain cell populations, or tissues outside the brain. Through a search for CLA-specific genes in mice, we identified *Smim32* (a gene with unknown function that is also enriched in the human CLA³⁷) as a promising candidate gene to drive recombinase expression in the CLA. Here we present the generation and characterization of different *Smim32* transgenic mouse lines expressing various recombinases in the CLA, providing a new toolbox for researchers aiming at studying the function of the enigmatic structure.

Results

***Smim32* is transcribed in the CLA, EN and TRN neurons**

We initially searched the entire Allen Mouse Brain Atlas database and visually selected genes with strong and relatively specific signals in the CLA. *Smim32* was found to be highly expressed in the CLA, as well as in the thalamic reticular nucleus (TRN). To precisely determine the identity of neurons expressing *Smim32* in the mouse brain, we re-analyzed published single-cell RNA sequencing (scRNAseq) datasets from dissections of the anterior part of the cortex and the thalamus³⁸[↗](#). Since the full-length mRNA of *Smim32* was not annotated at the time, the analysis involved re-mapping reads using an updated annotation of the mouse genome and reassignment of cellular identities through clustering and manual classification (see Materials and Methods). Among frontal cortex neurons, *Smim32* transcripts were highly enriched in CLA excitatory neurons (**Figure 1A-D**[↗](#)). These neurons formed a distinct cluster of transcriptomic identities characterized by the expression of *Nr4a2* and *Slc17a6* (**Figure 1B,D**[↗](#)). Among thalamic neurons, *Smim32* transcripts were highly enriched in inhibitory neurons of the TRN (**Figure 1E-H**[↗](#)). These neurons also formed a distinct cluster of transcriptome identities characterized by the expression of *Pvalb* (**Figure 1F,H**[↗](#)).

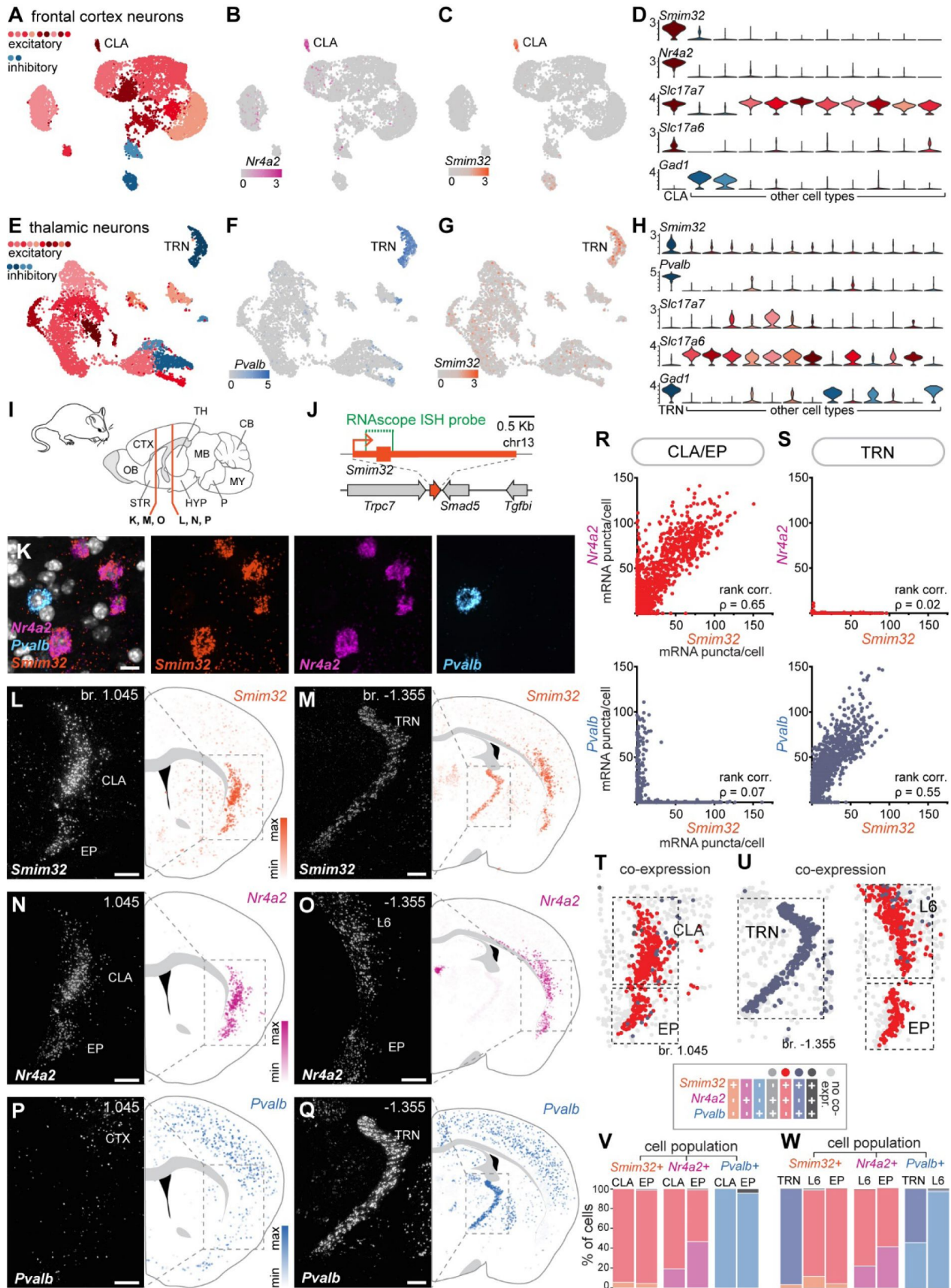


Figure 1

***Smim32* is expressed in the CLA, EP and TRN neurons.**

(A) UMAP plot of single-cell transcriptomes of neurons from the anterior third of the cortex (7804 cells). Excitatory and inhibitory cell types are colored with red and blue tones, respectively. Each cluster is represented by a different color. A single cluster could be identified as excitatory neurons pertaining to the CLA based on the expression of previously identified markers.

(B,C) Cells shaded according to the normalized expression of (B) *Nr4a2* and (C) *Smim32* across neurons of the anterior cortex.

(D) Density plots of normalized expression levels of selected genes across clusters.

(E) UMAP plot of single-cell transcriptomes of neurons from the thalamus (7317 cells) with a similar color coding as in A.

(F,G) Cells shaded according to the normalized expression of (F) *Pvalb* and (G) *Smim32* across neurons of the anterior cortex.

(H) Density plots of normalized expression levels of selected genes across clusters.

(I) Schematic of the localization of coronal sections shown in (L-Q). Anterior sections (K,M,O) comprise the CLA region, while posterior sections (L,N,P) include the TRN.

(J) Transcript topology (corresponding to the RefSeq NM_001378296.1) and genomic localization of *Smim32*. The region targeted by RNAscope ISH probes is highlighted in green.

(K) Illustrative image of CLA cells labeled by RNAscope ISH with probes for *Smim32* (orange), *Nr4a2* (magenta) and *Pvalb* (blue). Individual colored spots (mRNA puncta) correspond to mRNA transcripts. DAPI staining is shown in grey. Scale bar: 10 μ m.

(L,M) Left, representative images of RNAscope ISH labeling of *Smim32* transcripts in the (I) CLA/EP and in the (M) TRN. Scale bar: 250 μ m. Right, cells expressing *Smim32* across the entire hemisection, where dot color corresponds to the *Smim32*-expression level in each individual cell.

(N,O) Same as above, but showing *Nr4a2* expression in the CLA/EP and in cortical layer 6 (L6). Scale bar: 250 μ m.

(P-Q) Same as above, but showing *Pvalb* expression around the CLA/EP and in the TRN. Scale bar: 250 μ m.

(R,S) Correlation between expression levels of *Smim32* and *Nr4a2*, and *Smim32* and *Pvalb*, in cells from the (R) CLA/EP and cells from the (S) TRN. Each dot corresponds to one cell. All cells from both hemispheres, within the regions of interest (ROI, grey dotted line in (L,Q)), were included in the analysis. p value indicates Spearman's rank correlation. CLA/EP, n = 6755 cells; TRN, n = 11139 cells.

(T,U) Illustrative map of cells co-expressing *Smim32* and *Nr4a2* (red) or *Smim32* and *Pvalb* (purple grey) in the CLA/EP, the TRN, and in L6. A cell was considered positive for a marker when a minimum of 20 mRNA puncta are attributed to this cell. The dotted lines highlight the ROIs used for quantification analyses shown in (V,W).

(V,W) Bar plots showing the percentages of cells positive for one or several markers. The analysis was done separately for each cell population.

To visualize the expression pattern of *Smim32* in the brain, we labeled *Smim32* transcripts using quantitative RNA in situ hybridization (RNAscope ISH) (Figure 1I-K). Consistent with the scRNAseq analysis, the expression pattern of *Smim32* resembled that of the CLA marker *Nr4a2* within the CLA, EP, and cortical L6 (Figure 1L-O), and of *Pvalb* within the TRN (Figure 1P-Q). To determine the overlap of *Smim32* expression with known CLA and TRN markers, we quantified the expression levels of each marker in individual cells labeled by RNAscope ISH. We confirmed that *Smim32* expression levels correlated with expression levels of *Nr4a2* and *Slc17a6* in the CLA and EP, and with *Pvalb* and *Gad1* in the TRN (Figure 1R-W, Supplementary Figure 1). In the CLA, 94% of all *Smim32* positive cells co-expressed *Nr4a2*, while about 80% of all *Nr4a2* positive cells within the CLA co-expressed *Smim32* (Figure 1V). In the TRN, 96% of all *Smim32*-positive cells co-expressed *Pvalb*, and 54% of all *Pvalb*-positive cells co-expressed *Smim32* (Figure 1W). Thus, *Smim32* is expressed in a large majority of CLA excitatory neurons, as well as in a significant proportion of TRN inhibitory neurons.

***Smim32* is co-expressed with known CLA markers**

We further characterized the expression pattern of *Smim32* within the CLA by examining its potential co-expression with various known markers of the CLA and surrounding insular cortex. RNAscope ISH showed that *Smim32*-expression levels correlated with the expression of *Lxn*, *Gnb4*, *Oprk1*, and *Car3*, all of which are known markers of the CLA (Figure 2A-L). However, *Smim32* was not expressed in cell populations characterized by expression of *Ctgf* (also termed *Ccn2*, cortical layer 6a; Figure 2M-O) or *Rprm* (cortical layer 6b and piriform cortex; Figure 2P-R). Thus, we establish that *Smim32* is a highly specific marker of cells from the CLA and EP in mice.

Generation of mouse lines expressing recombinases in *Smim32*-neurons

To generate mice expressing recombinases in post-mitotic CLA excitatory neurons, we took advantage of the *Smim32* promoter in two separate transgenic approaches. On the one hand, we generated two knockin mouse lines of *Smim32* into which we inserted an IRES-recombinase (either Cre or Flippase) cassette after its endogenous stop codon (Figure 3A). These resulting alleles are later referred to as *Smim32*-Cre and *Smim32*-Flpo. To assess the specificity of these tools, we crossed these mice with recombinase-inducible reporter lines. *Smim32*-Cre mice were crossed with the Cre-inducible tdTomato-expressing reporter line Ai14³⁹, and *Smim32*-Flpo mice were crossed with the Flippase-inducible tdTomato-expressing reporter line Ai65F³⁹. Since reporter expression is driven by a CAG promoter and inserted in the Rosa26 locus in both the Ai14 and the Ai65F line, they are both further referred to as R26-tdT. Reporter expression in *Smim32*^{Cre/wt}; R26^{tdT/wt} adult mouse brain was found in a large proportion of the CLA and TRN (Figure 3B-C). Additionally, we observed reporter expression in sparse neurons of the caudoputamen region (Figure 3B) and mid-thalamic region (Figure 3C). The reporter expression pattern was similar in *Smim32*^{Flpo/wt}; R26^{tdT/wt} mouse (Figure 3D-E), except that fewer cells were observed in the mid-thalamic region (Figure 3E).

On the other hand, we took advantage of the potential variability in expression between mouse lines associated with classical transgenesis and random transgene insertion in the genome. To this aim, we produced a Cre-expressing transgene under the control of the putative *Smim32* promoter (Tg(*Smim32*-Cre)) by modifying a BAC containing a genomic fragment spanning the *Smim32* gene (Figure 3F). Successful insertions of this transgene led to different expression patterns observed using the same Cre-inducible reporter as for the *Smim32*-Cre line (R26^{tdT}). Two Tg(*Smim32*-Cre) lines were selected for their specific expression patterns and were further characterized in this work: Tg(*Smim32*-Cre)61Irod, whose Cre expression is restricted to the CLA (Figure 3G,H), and Tg(*Smim32*-Cre)62Irod, whose Cre expression is very similar to that of *Smim32*-Cre, with the exception of a lack of expression in the caudoputamen region (Figure 3I,J).

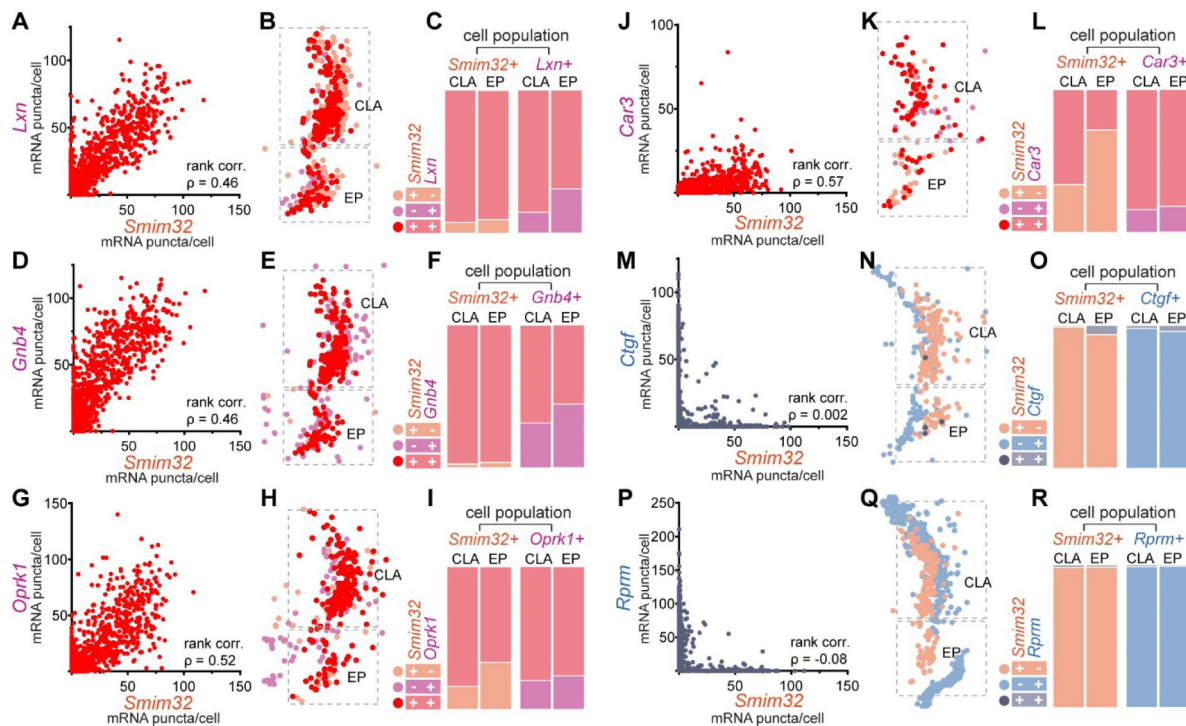


Figure 2

***Smim32* is co-expressed with known CLA markers.**

(A) Correlation between expression levels of *Smim32* and *Lxn* in cells from the CLA and EP. Each dot corresponds to one cell. *Smim32* and *Lxn*: ρ value indicates Spearman's rank correlation. $n = 6429$ cells. (B) Illustrative plot of cells co-expressing *Smim32* and *Lxn* (red) in the CLA and EP. Dotted lines highlight the ROIs used for quantification analyses shown in (C). (C) Bar plots showing the percentage of cells positive for one or both markers in the CLA and EP regions. (D–R) Same as above, but with different markers of the CLA and surrounding cell populations. (D–F) *Smim32* and *Gnb4*, $n = 6429$ cells. (G–I) *Smim32* and *Oprk1*, $n = 6780$ pairs. (J–L) *Smim32* and *Car3*, $n = 6493$ cells. (M–O) *Smim32* and *Ctgf*, $n = 7033$ cells. (P–R) *Smim32* and *Rprm*, $n = 6683$ cells.

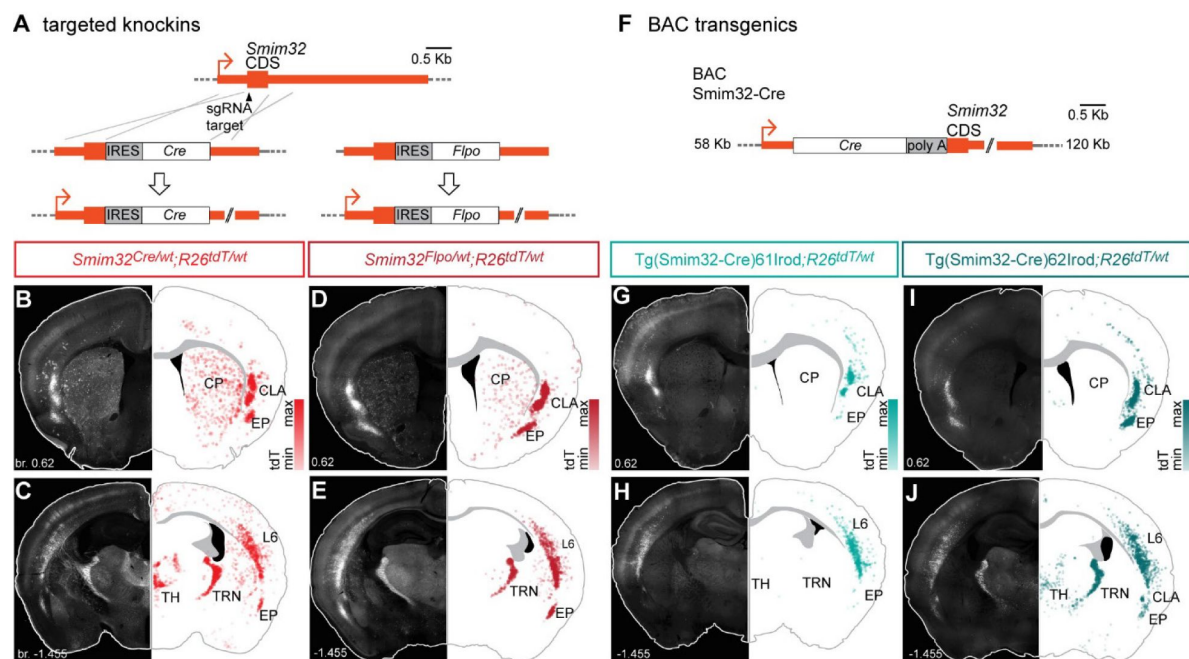


Figure 3

Generation of mouse lines expressing Cre in *Smim32*-neurons.

(A) Schematic of the CRISPR-mediated insertion of IRES-Cre or IRES-FlpO after the *Smim32* stop codon via homology direct repair (HDR) to generate the *Smim32*^{Cre} or the *Smim32*^{FlpO} alleles.

(B-E) Left, epifluorescence images of knockin mice crossed with mice carrying a Cre-dependent or a Flippase-dependent reporter allele *R26*^{tdT}. Representative coronal sections at the level of the CLA and TRN. Right, maps of tdT relative expression levels within the represented section. Each dot represents a cell.

(F) Schematic of the linearized BAC transgene use to generate the different Tg(*Smim32*-Cre) lines. A detailed description of the transgene can be found in the Materials and Methods section.

(G-J) Left, epifluorescence images of BAC transgenic mice (G,H: line 61Irod, I,J: line 62Irod) crossed with mice carrying a Cre-dependent reporter allele *R26*^{tdT}. Representative coronal sections at the level of the CLA and TRN. Right, maps of tdT relative expression within the represented section.

Characterization of the expression of reporter genes in adult mice

We then assessed the extent to which transgene-driven reporter expression patterns recapitulated *Smim32*-expression in the knockin and BAC transgenic mouse lines. We quantified single-cell expression of *Smim32* and *tdT* by RNAscope ISH in multiple sections covering different brain areas and at different anteroposterior levels (**Figure 4C**). In *Smim32^{Cre/wt}; R26^{tdT/wt}* mice, co-expression of *tdT* with *Smim32* reached up to 99.9% in parts of the CLA, indicating highly efficient recombination in these cells and biallelic expression of *Smim32* (**Figure 4F**). Interestingly, in *Tg(Smim32-Cre)61Irod; R26^{tdT/wt}* mice, *tdT* expression was absent in a large fraction of *Smim32*-positive neurons, particularly in the EP. In the TRN of the same line, transgene expression was almost null (**Figure 4G**), while being expressed in 61-81% *Smim32*-positive cells in the CLA. *Tg(Smim32-Cre)62Irod; R26^{tdT/wt}* mice showed a similar reporter expression pattern as *Smim32^{Cre/wt}; R26^{tdT/wt}* mice (**Figure 4H**). To evaluate reporter expression in the entire CLA and EP neuronal populations, we examined co-expression of the *tdT* reporter with *Nr4a2* (**Supplementary Figure 2**). Co-expression between *tdT* and *Nr4a2* reflected the co-expression of endogenous *Smim32* with *Nr4a2* (**Figure 1V,W**). In the CLA, *Tg(Smim32-Cre)61Irod; R26^{tdT/wt}* and *Tg(Smim32-Cre)62Irod; R26^{tdT/wt}* mice expressed *tdT* in 60-80% of all *Nr4a2*-positive cells, while in the EP mean co-expression was 23% and 43%, respectively.

To further assess the tissue specificity of the two transgenic lines, reporter expression was screened across all organs and tissues of newborn mice of the three Cre-expressing lines. Very sparse reporter-expressing cells were observed outside of the brain, likely resulting from leaky expression of the Cre, with the exception of the kidney medulla and papilla, where reporter-expressing cells were more abundant (**Supplementary Figure 3A-C**). To evaluate whether these cells expressed *Smim32* in a sustained manner, we re-analysed a published scRNAseq dataset of the mouse collecting ducts⁴⁰. In this dataset, *Smim32* transcripts were detected in no more than 5.2% of each subpopulation (**Supplementary Figure 3D**), suggesting that the observed patches of reporter-expressing cells result from leaky expression of the Cre and proliferation of recombined cells.

Characterization of transgenes' expression during development

To characterize Cre expression patterns of the different Cre lines in space and time, we first evaluated R26-tdT reporter expression in *Smim32^{Cre/wt}; R26^{tdT/wt}* mice between embryonic day 16 (E16) to postnatal day 200 (P200). The entire brains were sectioned and analyzed, and quantification was carried out in a representative set of coronal levels covering all regions where reporter-expressing cells were found in adult mice (**Figure 5A,D**, **Supplementary Figure 4A,B**). For mice aged from E16 to P8, we made a qualitative assessment of *tdT* expression onset (from when the bilateral expression was consistently observed in multiple cells on several consecutive sections). For later ages (P10-P200), we followed the number of reporter-expressing cells across the different regions and time points. Expression started in TRN, thalamus (TH), and main olfactory bulb (MOB), with the first labeled cells visible at E16. In the CLA and EP, expression started at P4. The proportion of cells expressing the reporter was highest in CLA, EP, TRN, and MOB in adult mice, while sparse expression was observed in TH, cortical layers 1-6 (CTX), caudoputamen (CP), and pons. For BAC transgenic mice, we evaluated reporter expression in mice from P0-P70 (**Figure 5B-C**, **Supplementary Figure 4C,D**). Expression of transgenes started later compared to the endogenous *Smim32* expression, beginning between P6-P8. Brain-wide analysis of reporter expression confirmed that expression of Cre in *Tg(Smim32-Cre)61Irod; R26^{tdT/wt}* mice is highly specific to the CLA, except for sparse expression in the EP and in CTX. In the latter, most of the labeled cells pertained to the layer 6 within posterior sections (see **Figure 3G** and **H** for an example).

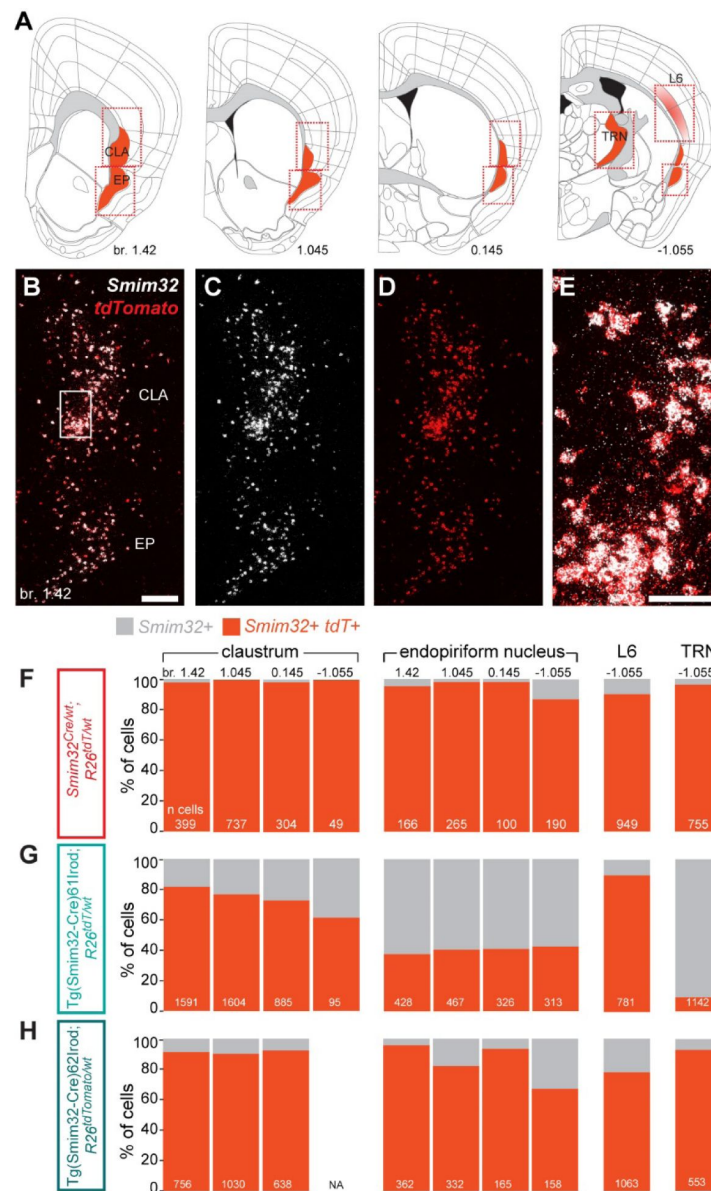


Figure 4

Characterization of the expression of reporter genes in adult mice.

(A) Schematics showing the regions (according to the Allen Brain Reference Atlases – Adult Mouse, atlas.brain-map.org) that were analyzed to compare expression levels of reporter genes in the knockin and in the BAC transgenic lines. Red dotted lines highlight the ROIs used for quantitative analyses shown in (F–H). br: distance from bregma in mm.

(B–D) RNAscope ISH labeling of *Smim32* (white) and *tdT* (red) mRNA on a coronal section in the anterior CLA of a *Smim32^{Cre/wt}; R26^{tdT/wt}* mouse. The white square highlights the region magnified in (E). Scale bar: 200 μ m.

(E) CLA cells co-expressing *Smim32* and *tdT*. Scale bar: 50 μ m.

(F–H) Bar plots showing the percentage of cells co-expressing *Smim32* with the Cre-inducible reporter allele *R26^{tdT}*. A cell was considered as positive for a marker when a minimum of 20 mRNA puncta were attributed to this cell. Numbers in the bar plots indicate the number of cells analyzed in the corresponding region and antero-posterior position.

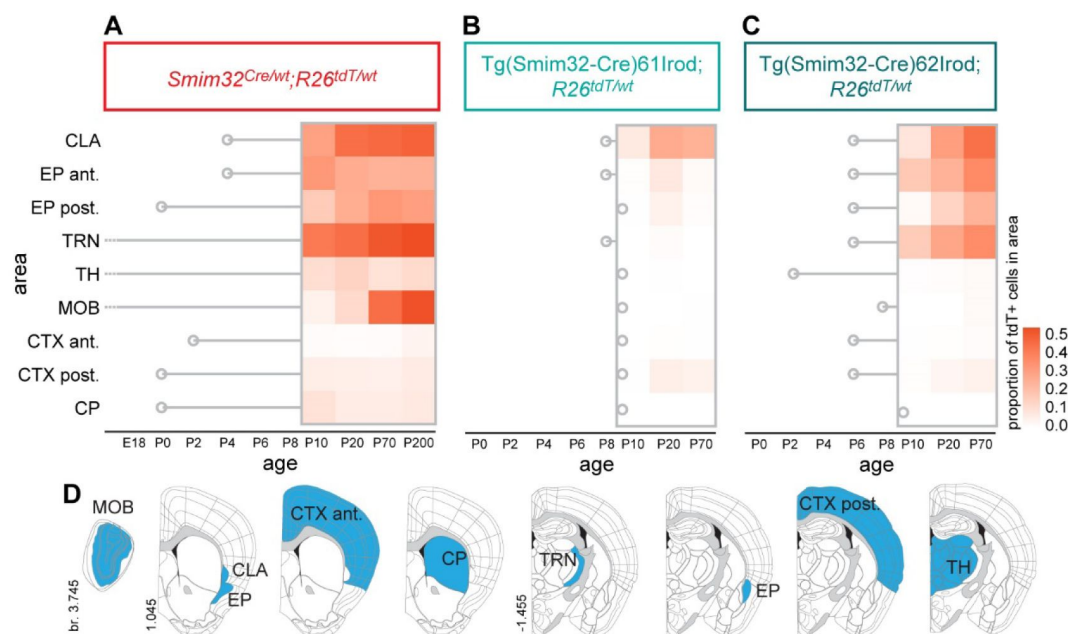


Figure 5

Expression of transgenes during development

(A-C) Expression of the Cre-dependent reporter gene *R26-tdT* in brain sections of (A) *Smim32-Cre*, (B) *Tg(Smim32-Cre)61Irod* and (C) *Tg(Smim32-Cre)62Irod* mice carrying the *R26^{tdT}* allele, from P10 to P200. Only brain regions with persistent expression in adult *Smim32-Cre* mice are shown here. A vertical grey bar indicates the first observation of *tdT*-positive cells within the structure. The heatmap shows relative gene expression levels in sections from P10, normalized across all regions and all mouse lines.

(D) Schematics of the regions of interest (blue) where *tdT* expression was quantified.

Cre-inducible tools used with *Smim32* BAC-transgenics

To validate the versatility of the different transgenic Cre-lines generated in this study, we combined our lines with various Cre-inducible genetic tools. Firstly, to test AAV-mediated gene delivery in CLA neurons, we stereotactically injected a Cre-dependent ChR2-EYFP-expressing AAV2 vector in the CLA region of a Tg(Smim32-Cre)61Irod adult mouse. Three weeks later, immunolabeling of YFP showed restricted recombination in neurons of the CLA (**Figure 6A,B**). Next, to assess the expression of a different Rosa26 knockin Cre-dependent allele than *R26-tdT*, Tg(Smim32-Cre)61Irod and Tg(Smim32-Cre)62Irod mice were each crossed with the *Gt(ROSA)26Sor^{tm2}(CAG-CHRM3*,-mCitrine)^{Ute}/J* (*R26-HA-hM3Dq*) line, which expresses the designer receptor exclusively activated by designer drugs (DREADD) hM3Dq upon Cre recombination (**Figure 6C-H**). When driven by Tg(Smim32-Cre)61Irod, hM3Dq expression was restricted in CLA excitatory neurons, along with very sparse cells in the caudoputamen and the EP (**Figure 6C-D**). Furthermore, we evaluated the feasibility of depleting CLA excitatory neurons by crossing Tg(Smim32-Cre)61Irod with the *NSE^{DTA/wt}* line, a Cre-inducible diphtheria toxin-expressing reporter transgenic line. Tg(Smim32-Cre)61Irod; *NSE^{DTA/wt}* mice displayed an average reduction of 49% of CLA Nr4a2-expressing neurons (Figure I-O). We also examined the expression patterns following BAC transgenic Cre-driven recombination of another family of Cre-inducible alleles, namely Ai155 (TIGRE-Optopatch3) and Ai148D (TIGRE-GCaMP6f). Expression of these reporter genes is controlled by a Tet-responsive element, which relies on a Cre-inducible Tet-transactivator, both of them inserted in the *Igs7* (TIGRE) locus. Tg(Smim32-Cre)61Irod; *TIGRE^{Optopatch3}/wt* mice displayed specific expression of the Optopatch3 complex in CLA neurons (**Figure 6P,Q**), and Tg(Smim32-Cre)62Irod; *TIGRE^{GCaMP6f}/wt* displayed specific expression of GCaMP6f in the CLA and TRN neurons (**Figure 6R,S**). Both reporter expression patterns, respective to the inducing line, are congruent with that of the *R26-tdT* line under the same condition. In conclusion, the *Smim32*-driven recombinase-expressing lines offer versatile possibilities and reproducibility across different Cre-inducible tools.

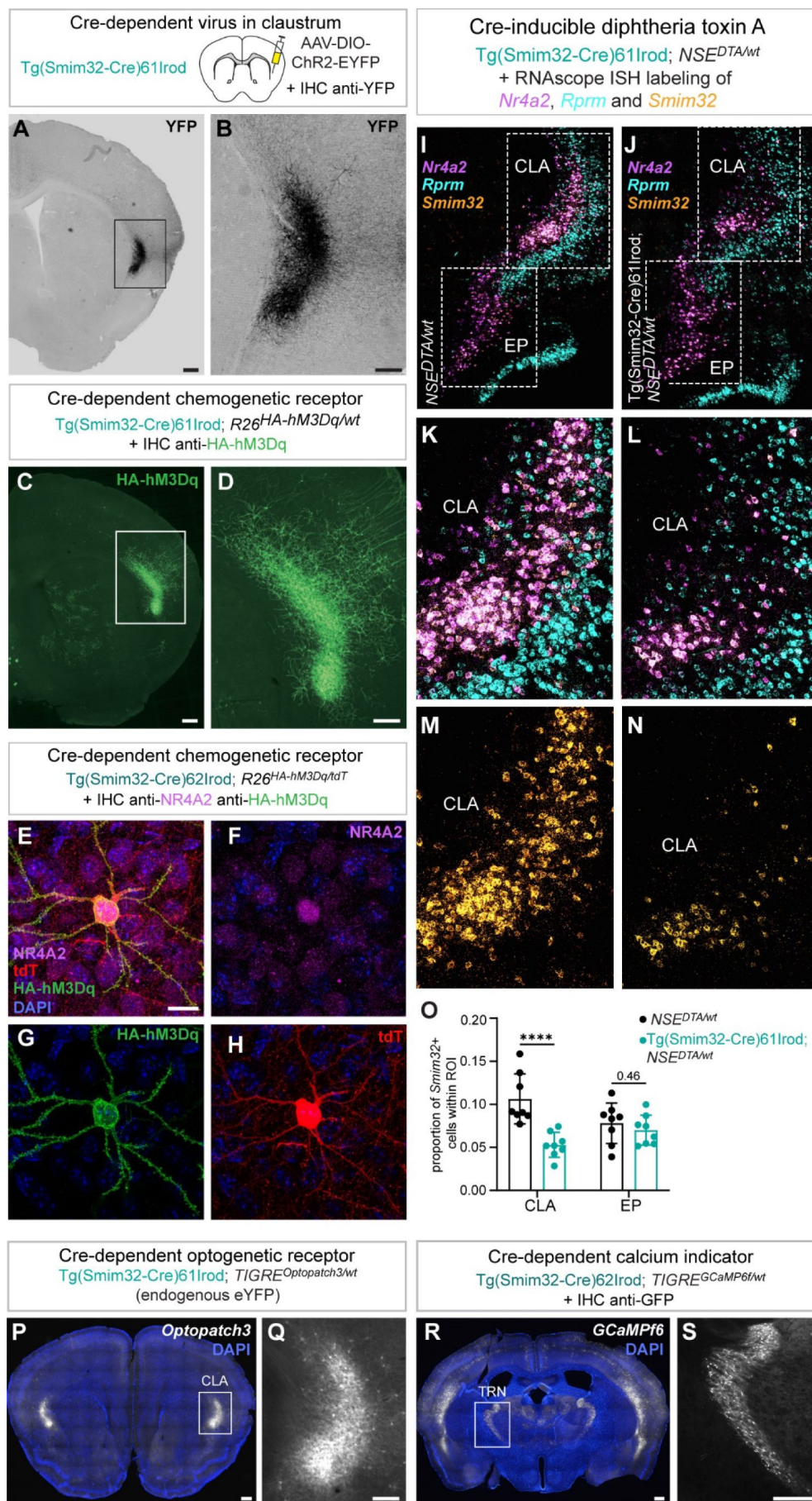


Figure 6

Examples of Cre-dependent reporters used with *Smim32* BAC transgenics.

(A,B) Tg(*Smim32*-Cre)61Irod mouse expressing ChR2 and EYFP after injection of a Cre-dependent viral vector with ChR2 and EYFP in the CLA. DAB IHC staining of YFP in a coronal section. The black square indicates the region magnified in (B). Scale bars: 200 μ m (A) and 100 μ m (B).

(C,D) Coronal section of a Tg(*Smim32*-Cre)61Irod; *R26^{HA-hM3Dq/wt}* mouse, where the HA-hM3Dq tag was labeled by IHC (green). The white square highlights the region magnified in (D). Scale bars: 200 μ m (C) and 100 μ m (D).

(E-H) Confocal image of a pyramidal neuron in the CLA periphery of a Tg(*Smim32*-Cre)62Irod; *R26^{HA-hM3Dq/tdT}*, DAPI staining of the nuclei (blue), (F) NR4A2 immunolabelling (magenta), (G) HA immunolabelling (green) and (H) tdT endogenous fluorescence (red). Scale bar: 15 μ m.

(I,J) Expression of the Cre-inducible diphtheria toxin A in *Smim32*-positive cells of the CLA. Tg(*Smim32*-Cre)61Irod mice were crossed with *NSE^{DTA/wt}* (neuron-specific enolase 2) mice, and the persistence of *Smim32*-positive cells in the CLA and EP was evaluated by RNAscope ISH labeling. Coronal sections of a Tg(*Smim32*-Cre)61Irod; *NSE^{DTA/wt}* and a control *NSE^{DTA/wt}* mouse were labeled with probes against *Smim32* (yellow), *Nr4a2* (magenta) and the cortical layer 6 marker *Rprm* (light blue).

(K-N) Higher magnification of the CLA region, with all three channels (K,L) and only the *Smim32* channel (M,N).

(O) Proportion of *Smim32*-positive cells within the CLA and EP (among all cells in the ROI, dotted lines in (I,J)). Tg(*Smim32*-Cre)61Irod; *NSE^{DTA/wt}*, n=1 mouse, n= 8 hemisections; *NSE^{DTA/wt}*, n=1 mouse, n= 8 hemisections. **** $p < 0.001$, two-way RM ANOVA, Fisher's LSD test.

(P,Q) Coronal section of a Tg(*Smim32*-Cre)61Irod; *TIGRE^{Optopatch3^{Cre}/wt}* mouse, with endogenous eYFP fluorescence. DAPI staining of nuclei (blue). The white square highlights the region magnified in (N). Scale bars: 200 μ m (M) and 100 μ m (N).

(R,S) Coronal section of a Tg(*Smim32*-Cre)62Irod; *TIGRE^{GCaMP6^{Cre}/wt}* mouse, where GCaMP6 is revealed by IHC labelling of GFP. DAPI staining of nuclei (blue). Scale bar: 200 μ m.

Discussion

While many groups have attempted to uncover the functions of the CLA, most approaches used to date to target and manipulate the mouse CLA have suffered from an unsatisfactory control or poor description of the cell types studied. In the mouse, neurons with a genetic identity of CLA neurons are nestled in, and even intermingled with other cell populations characterized by the expression of markers such as *Ctgf*, *Nnat*, *Nos1ap* or *Rprm*¹⁵. Hence, when targeting the CLA by viral injections without the involvement of cell type-specific recombinases, infection of cells from neighboring cell populations is inevitable. This problem cannot be circumvented with dual viral strategies, such as, for instance, the labeling of a subpopulation of CLA projection neurons based on the target area, as the CLA specificity of this approach is not established. Moreover, the use of viruses is often unsatisfactory as the CLA neuronal population is never targeted in its entirety. Unsurprisingly, many approaches essentially based on virus injections have led to apparently contradictory findings and have slowed the emergence of a consensus relative to the various dimensions of the CLA, ranging from its projection pattern to its function.

In this work, we designed a series of recombinase-expressing mouse lines leveraging the promoter of *Smim32*, a gene of unknown function whose expression is specific to CLA and EN projection neurons, as well as to inhibitory neurons of the TRN. These novel *Smim32-Cre* and *Smim32-FlpO* alleles provide tools, based on the unique expression pattern of *Smim32*, to specifically target the CLA and EN in the claustrinsular region.

Moreover, taking advantage of peculiar expression patterns resulting from specific integration sites of transgenes, we also generated two additional transgenic lines, *Tg(Smim32-Cre)61Irod* and *Tg(Smim32-Cre)62Irod* BAC, which exhibit a more restrictive transcription pattern than the knockin alleles, in particular the lack of expression in the olfactory bulb.

The major advantage of the *Tg(Smim32-Cre)61Irod* mouse line is its high specificity to CLA neurons: the transgene is expressed on average in 70% of CLA projection neurons (*Nr4a2*+ population), while being only sparsely expressed in the EP and other cell populations of the claustrinsular region. Compared to previously described transgenic mouse lines, such as *Egr2-Cre*¹⁷, *Tg(Tbx21-Cre)*¹⁸, or *Gnb4-IRES2-CreERT2*³⁶ mice, which are either expressed in only a small fraction of CLA neurons, or which suffer from high expression in the EP, this mouse allows the specific targeting of a large fraction of CLA neurons.

Taking advantage of the new *Smim32-FlpO* allele, even more specific targeting of CLA projection neurons could be achieved via an intersectional approach, involving, for instance, *Slc17a6-Cre* and *Smim32-FlpO*. Using the desired reporter, this will allow broad characterization of CLA projections, specific FACsorting of CLA neurons, optogenetic and pharmacogenetic manipulation of the whole structure, without any virus delivery.

With these genetic tools in hand, the comprehensive targeting and functional probing of the densely connected CLA is now possible.

Materials and methods

Animals

All experiments were conducted in accordance with the veterinary guidelines and regulations of the University and of the state of Geneva (Direction of Animal Experimentation, UNIGE). C57BL/6J male mice were purchased at 5–7 weeks of age from Charles River Laboratories. Upon arrival, they were group-housed in standard type II cages with *ad libitum* access to food and water. They were held on a dark/light cycle of 12/12 h, temperature between 21 and 22 °C, and humidity between 45 and 55%.

Generation of transgenic mice

BAC transgenics *Tg(Smim32-cre)61Irod* and *Tg(Smim32-cre)62Irod*

This transgene was obtained by a modification of the bacterial artificial chromosome (BAC) RP23-125G21, from the RPCI-23 BAC library⁴¹. This BAC spans from the 3' intergenic region in 5' of *Smad5* to the second intron of *Trpc7*, therefore encompassing *Smim32*, which is located in between these two genes. To generate this transgene, we inserted the cassette DDCre-Rabbit β -globin polyA-frr-PuroR-ZeoR-frr in place of *Smim32* start codon via recombineering. The resulting vector was injected in the pronuclei of C57BL/6J-DBA/2J F2 zygotes. Four founders were obtained out of which two lines were established for their expression pattern specificities, hereafter named *Tg(Smim32-cre)61Irod* and *Tg(Smim32-cre)62Irod*. DDCre refers to the fusion of a destabilizing tag at the N-terminus of the Cre recombinase, targeting the protein to proteasome degradation⁴², a process that is inhibited by trimethoprim (TMP) treatment. Three intraperitoneal injections of 300 μ g of

TMP/gram of body weight on three consecutive days one week prior to analysis (one-month-old animals) did not affect the reporter (R26-tdT) expression pattern in either transgenic line. For this reason, we removed the DD prefix from our nomenclature.

Smim32-Cre

This knockin was obtained via CRISPR-induced homology direct DNA repair (HDR) using a circular donor template. The vector of this template is a pUC57 plasmid in which we inserted an IRES-Cre (Cre) cassette flanked by homology arms corresponding to 801 bp and 799 bp before and after the cut site, respectively. The expected cut site, which is also the insertion site, was located on the *Smim32* stop codon. The HDR donor along with the Cas9 complex (Cas9NLS protein, tracrRNA and crRNA) was injected in the pronuclei of C57BL6/J mice. A unique founder was selected and backcross for at least three generations before producing the mice analyzed in this study.

The guide RNA target sequence is AGACGCGACCGGTATAGGGC(AGG) (PAM sequence in parentheses).

Smim32-Flpo

This knockin was obtained via CRISPR-induced HDR using a Tild vector⁴³. The guide RNA target is the same as for Smim32-Cre. The targeting vector was modified from used for Smim32-Cre, with slightly shorter homology arms resulting from blunt digestion (796 bp and 797 bp in 5' and 3', respectively) and the codon-optimized Flippase (FlpO) coding sequence in place of the Cre. The donor template (a linear fragment composed of the aforementioned sequences) was injected along with the Cas9 complex in the pronuclei of C57BL6/J mice. A unique founder was selected and backcross for at least two generations before producing the mouse analyzed in this study.

Genotyping

Tg(Smim32-cre) mice can be genotyped by PCR with the primers #1 and #2 (see **Table 1**), yielding a 854 bp amplicon from the transgenic allele. *Smim32-Cre* mice can be genotyped with a mix of the primers #3, #4 and #5, yielding a 400 bp amplicon from the *Smim32-Cre* allele and a 645 bp amplicon from the wild-type allele. *Smim32-Flpo* mice can be genotyped with a mix of the primers #3, #5 and #6, yielding a 423 bp amplicon from the *Smim32-Flpo* allele and a 645 bp amplicon from the wild-type allele.

Reporter Mouse lines

Details regarding all transgenic mouse lines used are provided in **Table 2**.

Single-cell RNAseq analysis

Brain

Single-cell RNA-sequencing (scRNA-seq) datasets from the Linnarson lab's mouse brain atlas (<http://www.mousebrain.org>) corresponding to the anterior part of the cortex and to the thalamus were downloaded from NCBI's sequence read archive (SRA, see run accessions in **Table 3**).

Smim32 3'UTR was partially annotated at the time of the publication of Linnarson lab's mouse brain atlas³⁸ and, therefore not detected. To capture *Smim32* expression levels in these data, we updated the Ensembl annotation version 86 of the mouse genome assembly mm10/GRCm38 with the annotation of the RefSeq mRNA NM_001378296.1, which exhaustively covers the 3' UTR. Extraction of the raw data and mappings on the updated mouse genome annotation were performed with the cellranger software suite, provided by the 10X Genomics website (<https://>

Table 1

Primers for genotyping.

#	target	Sequence
1	Fwd in <i>Smim32</i> 5' UTR	CTGGAAAGCCGTAAAACGGG
2	Rev in Cre CDS	ATCGACCGGTAATGCAGGCAA
3	Fwd in <i>Smim32</i> CDS	ATGTACGGCGACGTGTTCAA
4	Fwd in Cre CDS	CCGTAACCTGGATAGTGAAACAGG
5	Rev in <i>Smim32</i> 3' UTR	CAGCTCCTGATTAGTCCCGC
6	Fwd in FlpO CDS	CATCAGATACCCCGCCTGGA

Table 2

List of transgenic mouse lines.

Figures	Name in paper	Official name	Reporter reference	mouse	RRID
3(B,C), (G-J), 4(B-H), Supp. Fig. 2(A-C), Supp. Fig. 3(B,C), 5(A-C), Supp. Fig. 4(B-D), 6(E,H)	R26-tdT	B6;129S6- <i>Gt(ROSA)26Sor^{tm14}(CAG-tdTomato)Hze/J</i> , Ai14	Madisen et al. 2010, PubMed:20023653		IMSR_JAX :007908
3(D,E)	R26-tdT	B6;129S- <i>Gt(ROSA)26Sor^{tm65.2}(CAG-tdTomato)Hze/J</i>	Madisen et al. 2010, PubMed:20023653		IMSR_JAX :032864
6(C-H)	R26-HA-hM3Dq	B6N;129- <i>Gt(ROSA)26Sor^{tm2}(CAG-CHRM3*,mCitrine)Ute/J</i>	Zhu et al. 2016, PubMed:27194399		IMSR_JAX :026220
6(I-O)	NSE-DTA	<i>Eno2^{tm1}(DTA)Hsak</i>	17989651 Kobayakawa		MGI: 3765961
6(P,Q)	TIGRE- Optopatch3	B6;129S6- <i>Igs7^{tm3}(tetO-Optopatch3,CAG-tTA)Acoh/J</i> , Ai155	PubMed:30952812 , Farhi et al. 2019		IMSR_JAX :029679
6(R,S)	TIGRE- GCaMP6f	B6.Cg- <i>Igs7^{tm148.1}(tetO-GCaMP6f,CAG-tTA2)Hze/J</i> , Ai148D	-		IMSR_JAX :030328

<i>sample identifier</i>	<i>tissue</i>	<i>SRA run accession</i>
10X05_1	anterior cortex	SRR6854100
10X05_2	anterior cortex	SRR6854145
10X22_1	anterior cortex	SRR6854092
10X22_2	anterior cortex	SRR6854170
10X24_3	thalamus	SRR6854165
10X24_4	thalamus	SRR6854164
10X50_5	thalamus	SRR6854143
10X50_6	thalamus	SRR6854144
10X50_7	thalamus	SRR6854137
10X50_8	thalamus	SRR6854138
10X57_6	thalamus	SRR6854074
10X57_7	thalamus	SRR6854064

Table 3

SRA run accessions for single-cells RNAseq analysis.

support.10xgenomics.com/). Sequences were extracted from the BAM files using the program `bamtofastq`, and mapping and counting were performed with the count program, that was run using the options `--force-cells=3500`. The clustering analysis was performed using the Seurat package⁴⁴. First, single cell transcriptomes (thalamus: 37000 cells from height samples, anterior cortex: 14000 cells from four samples) were filtered to remove datasets with less than 600 counts or with an insufficient ratio of UMI per number of genes (nGene) detected ($\text{UMI/nGene} \geq 1.2$). Next, genes that were present in less than 20 cells were removed from the analysis. The samples were merged and integrated using the SCTransform protocol, available on the Seurat website (<https://satijalab.org/seurat/v3.1/integration.html>). Clusters were formed using the `findneighbours` and `findcluster` functions from Seurat with the principal components (anterior cortex: PC 1-8, resolution 0.2, thalamus: PC 1-14, resolution 0.2). Next, we reduced the dataset to neurons by selecting cells expressing the neuronal marker *Snap25*. A second integration was performed on this subset using the same protocol as for the whole dataset, but with slightly different parameters (anterior cortex: PC 1-8, resolution 0.1, thalamus: PC 1-13, resolution 0.2). Finally, a dimensionality reduction was performed with the UMAP method to project the transcriptomes on a 2-dimensional plot (Figure1A-D).

Kidney

Mouse kidney collecting duct scRNAseq data was retrieved from NCBI's SRA under the accession SRP108643. Mapping on the mouse genome assembly mm10 was performed with STAR⁴⁵, restricting to uniquely mapped reads with the option `--outFilterMultimapNmax=1`. Quantification of *Smim32* expression was performed with `featureCounts`⁴⁶ using the same updated gene annotation file as for the brain scRNAseq data. Cell type annotation was retrieved from the corresponding GEO dataset GSE99701. To estimate the fraction of *Smim32*-expressing cells across cell types (pie charts in **Supplementary Figure 3D**), we counted the number cells with *Smim32* transcript levels reaching more than 1 TPM over the total number of cells.

RNAscope *in situ* hybridization

Mice were anesthetized with isoflurane and killed by cervical dislocation, brains were immediately embedded in OCT and frozen in an isopentane bath in liquid nitrogen. Samples were stored at -80°C until sectioning. Brains were cut on a cryostat microtome in $10\ \mu\text{m}$ coronal sections and mounted on Superfrost Plus slides. Sections were dried for 1-2 hours at -20°C inside the cryostat, before being transferred for storage at -80°C . Individual mRNA molecules on sections were labeled using the RNAscope™ Multiplex Fluorescent V2 Assay (ref. 323136, Advanced Cell Diagnostics), following the guidelines for fresh-frozen tissue. Before probe hybridization, sections were post-fixed in Formalin 10% for 15 min at 4°C , dehydrated, treated with hydrogen peroxide for 10 min at room temperature, and treated with Protease IV for 15 min at room temperature. Probes were visualized with TSA Vivid Fluorophores 520 (1:750), ref. 323271, 570 (1:1500) ref. 323272 and 650 (1:2000), ref. 323273, diluted in TSA buffer. Sections were counterstained with DAPI and mounted with ProLong™ Gold antifade (ref. P36935, Invitrogen). Slides were imaged with a Nikon Ti/CSU-W1 Spinning Disc Confocal microscope using 405, 488, 561 and 640 nm excitation lasers and a 60x 1.49 NA air objective. Image tiles of 6 Z planes cover $4\ \mu\text{m}$ were acquired, merging of tiles and orthogonal projection was applied within the SlideBook software (Intelligent Imaging Innovations).

RNAscope probes

All RNAscope probes used in the present study can be found in **Table 4**.

Table 4

List of RNAscope probes

Figure	Sections	Mouse	Probes	Reference
1(K-W)	CLA, bregma 1.045, n=2 TRN, bregma -1.355, n=2	C57BL6J n=1	<i>Smim32</i> <i>Nr4a2</i> <i>Pvalb</i>	Mm-Gm45623-O2-C1, 1131201 Mm-Nr4a2-C2, 423351-C2 Mm-Pvalb-C3, 421931-C3
Supp. Fig. 1(A-F)	CLA, bregma 1.045, n=2 TRN, bregma -1.355, n=2	C57BL6J n=1	<i>Smim32</i> <i>Gad1</i> <i>Slc17a6</i>	Mm-Gm45623-O2-C1, 1131201 Mm-Gad1-C2, 4100951-C2, Mm-Slc17a6-C3, 319171-C3
2(A-F)	CLA, bregma 1.045, n=2	C57BL6J n=1	<i>Gnb4</i> <i>Smim32</i> <i>Lxn</i>	Mm-Gnb4, 460951 Mm-Gm45623-O2-C2, 1104471-C2 Mm-Lxn-C3, 585801-C3
2(G-I)	CLA, bregma 1.045, n=2	C57BL6J n=1	<i>Smim32</i> <i>Oprk1</i>	Mm-Gm45623-O2-C1, 1131201 Mm-Oprk1-C2, 316111-C2
2(J-L)	CLA, bregma 1.045, n=2	C57BL6J n=1	<i>Smim32</i> <i>Car3</i>	Mm-Gm45623-O2-C1, 1131201 Mm-Car3-C2, 578571-C2
2(M-O)	CLA, bregma 1.045, n=2	C57BL6J n=1	<i>Smim32</i> <i>Ctgf</i>	Mm-Gm45623-O2-C2, 1104471-C2 Mm-CTGF, 314541
2(P-R)	CLA, bregma 1.045, n=2	C57BL6J n=1	<i>Smim32</i> <i>Rprm</i>	Mm-Gm45623-O2-C1, 1131201 Mm-Rprm-C2, 466071-C2
4(A-H) Supp. Fig. 2(A-C)	<i>Smim32-Cre</i> CLA bregma 1.42, 0.145, -1.055 n=2, 1.045 n=3, TRN n=4; Tg(<i>Smim32-Cre</i>)61Irod; <i>R26^{tdT/wt}</i> CLA 1.42, 1.045, 0.145 n=8, -1.055 n=3, TRN n=4; Tg(<i>Smim32-Cre</i>)62Irod; <i>R26^{tdT/wt}</i> CLA 1.42, 1.045, 0.145 n=4, -1.055 n=3, TRN -1.055 n=4.	<i>Smim32^{Cre/wt}</i> , <i>R26^{tdT/wt}</i> n=1, Tg(<i>Smim32-Cre</i>)61Irod; <i>R26^{tdT/wt}</i> n=2, Tg(<i>Smim32-Cre</i>)62Irod; <i>R26^{tdT/wt}</i> n=1	<i>Smim32</i> <i>tdTomato</i> <i>Nr4a2</i>	Mm-Gm45623-O2-C1, 1131201 tdTomato-C2, 317041-C2 Mm-Nr4a2-C3, 423351-C3
6(I-O)	CLA, bregma 1.045, n=8	<i>NSE^{DTA/wt}</i> , n=1 Tg(<i>Smim32-Cre</i>)61Irod; <i>NSE^{DTA/wt}</i> , n=1	<i>Smim32</i> <i>Nr4a2</i> <i>Rprm</i>	Mm-Gm45623-O2-C1, 1131201 Mm-Nr4a2-C3, 423351-C3 Mm-Rprm-C2, 466071-C2

Figure	Sections	Mouse	Primary	Reference	Secondary	Reference
6(A,B)	CLA, bregma 1.045, n=2	Tg(Smim32- Cre)61Irod; AAV- DIO-ChR2-EYFP Female, n=1	Rabbit anti- YFP, 1:1000	Invitrogen, A6455	Biotinylated goat anti- rabbit IgG, 1:200	Vector laboratories, BA-1000
6(C,D)	CLA, bregma 1.045, n=2	Tg(Smim32- Cre)61Irod; <i>Rosa26^{HA-hM3Dq/wt}</i> Female, n=1	Rabbit anti HA tag, 1:1000	Cell Signaling, C29F4	Goat anti- rabbit Alexa 488 1:500	Life technologies, A11034
6(E-H)	CLA, bregma 1.045, n=2	Tg(Smim32- Cre)62Irod; <i>Rosa26^{HA-hM3Dq/tdTomato}</i> Male, n=1	Goat anti- NR4A2, 1:200, Rabbit anti- HA tag, 1:1000	R&D systems, Bio-technie, AF2156 Cell Signaling, C29F4	Donkey anti- goat Alexa 633 1:500 Donkey anti- rabbit Alexa 488 1:500	ThermoFisher Scientific, A-21082 Abcam, ab150073
6(P,Q)	CLA, bregma 1.045, n=2	Tg(Smim32- Cre)61Irod; <i>TIGRE^{Optopatch3/wt}</i> Male, n=1	Endogenous eYFP	-	-	-
6(R,S)	TRN, bregma -1.055, n=2	Tg(Smim32- Cre)62Irod; <i>TIGRE^{GCaMP6/wt}</i> Male, n=1	Rabbit anti- GFP, 1:500	Abcam, ab290	Donkey anti- rabbit Alexa 488, 1:500	Abcam, ab150073

Table 5

List of antibodies used for IHC labeling.

Figure	Parameter	Structure of data	Type of test	Sample size	Statistical data
1R	Number of <i>Nr4a2</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6755 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.6445$, 95% confidence interval = 0.6299 to 0.6586, $p < 0.0001$
1R	Number of <i>Pvalb</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6755 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.06525$, 95% confidence interval = 0.0408 to 0.0897, $p < 0.0001$
1S	Number of <i>Nr4a2</i> and <i>Smim32</i> mRNA puncta/cell in the TRN	Non normal distribution	Spearman's rank correlation	n= 11139 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.02402$, 95% confidence interval = 0.0049 to 0.0433, $p = 0.0112$
1S	Number of <i>Pvalb</i> and <i>Smim32</i> mRNA puncta/cell in the TRN	Non normal distribution	Spearman's rank correlation	n= 11139 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.5472$, 95% confidence interval = 0.5337 to 0.5605, $p < 0.0001$

Table 6

Statistical data.

Supp. Figure 1A	Number of <i>Slc17a6</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6272 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.3277$, 95% confidence interval = 0.3048 to 0.3503, $p < 0.0001$
Supp. Figure 1A	Number of <i>Gad1</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6272 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.0636$, 95% confidence interval = 0.0382 to 0.0889, $p < 0.0001$
Supp. Figure 1B	Number of <i>Slc17a6</i> and <i>Smim32</i> mRNA puncta/cell in the TRN	Non normal distribution	Spearman's rank correlation	n= 11101 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.0419$, 95% confidence interval = 0.0227 to 0.0610, $p < 0.0001$
Supp. Figure 1B	Number of <i>Gad1</i> and <i>Smim32</i> mRNA puncta/cell in the TRN	Non normal distribution	Spearman's rank correlation	n= 11101 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.6212$, 95% confidence interval = 0.6093 to 0.6328, $p < 0.0001$
2A	Number of <i>Lxn</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6429 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.4626$, 95% confidence interval = 0.4426 to 0.4821, $p < 0.0001$
2D	Number of <i>Gnb4</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6429 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.4629$, 95% confidence interval = 0.4429 to 0.4824, $p < 0.0001$
2G	Number of <i>Oprk1</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6780 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.5178$, 95% confidence interval = 0.4996 to 0.5355, $p < 0.0001$
2J	Number of <i>Car3</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6493 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.5678$, 95% confidence interval = 0.5505 to 0.5845, $p < 0.0001$
2M	Number of <i>Ctgf</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 7033 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = 0.0021$, 95% confidence interval = -0.0220 to 0.0262, $p = 0.8609$
2P	Number of <i>Rprm</i> and <i>Smim32</i> mRNA puncta/cell in the CLA and EP	Non normal distribution	Spearman's rank correlation	n= 6683 cells, 2 sections, 1 C57BL6/J male mouse	$\rho = -0.0796$, 95% confidence interval = -0.1041 to -0.0550, $p < 0.0001$
6O	Proportion of <i>Smim32</i> ⁺ cells within ROI	NA	Two-way ANOVA, Fishers LSD post hoc test	n= 8 sections from 1 <i>NSE^{DTA/wt}</i> male mouse, n=8 hemisections	$F_{\text{area} \times \text{genotype}(1,14)} = 28.18$, $p = 0.0001$, $F_{\text{area}(1,14)} = 1.685$, $p = 0.2153$

Table 6 (continued)

	from 1 Tg(Smim32- Cre)61Irod; <i>NSE^{DTA/wt}</i> mouse	$F_{\text{genotype}(1,14)} = 9.354$, $p = 0.0085$ $F_{\text{section}(14,14)} = 5.531$, $p = 0.0014$ Post hoc test: <i>NSE^{DTA/wt}</i> vs. Tg(Smim32- Cre)61Irod; <i>NSE^{DTA/wt}</i> , mean difference in CLA = 0.05342, $p < 0.0001$, Mean difference in EP = 0.008054, $p = 0.4670$
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Table 6 (continued)

RNAscope image analysis

Images were analyzed with QuPath, version 0.4.3. First, cell segmentation based on DAPI staining was performed using the *Cell detection* function. Nuclei with a total area below $30 \mu\text{m}^2$ were excluded from the analysis. To include transcripts located in the cytoplasm, the area of the cells was expanded by $2 \mu\text{m}$ around the nucleus. Next, puncta corresponding to individual mRNA molecules were automatically detected and quantified using the *Subcellular detection* function. For each probe/fluorophore, a detection threshold is set, and all puncta above this threshold are counted. The same threshold is applied to all images labeled with a specific probe. When multiple puncta are clustered together, an estimation of the number of individual puncta within the cluster is performed automatically based on the intensity and size of the cluster.

Co-expression analysis

Cells were considered positive for a gene expression when a minimum of 20 mRNA puncta of the corresponding probe were attributed to this cell. This threshold was optimized in order to exclude false positives. The threshold was adapted to a minimum of 5 puncta/cell when the overall gene expression was very low in the cell population (the case for *Car3* and *Slc17a6*).

Cre/FlpO reporter imaging and analysis

Mice heterozygous for *Smim32-Cre* or hemizygous for Tg(*Smim32-Cre*)61Irod or Tg(*Smim32-Cre*)62Irod were crossed with a reporter mouse line homozygous for *Rosa26-stop(LoxP)-tdTomato*, to generate heterozygous reporter mice. Expression of tdTomato in *Smim32-Cre* male and female mice were analyzed at age E16, E18, P0, P2, P4, P6, P8, P10, P12, P14, P16, P18, P20, P70 and P200. Tg(*Smim32-Cre*)61Irod and Tg(*Smim32-Cre*)62Irod male mice were analyzed at age P0, P2, P4, P6, P8, P10, P12, P14, P16, P18, P20 and P70. A male *Smim32-FlpO* mouse was analyzed at P20. One mouse per time point, sex and genotype were analyzed.

Embryos (E16-E18) and Newborns (P0-P8)

To collect embryos, pregnant mothers received an intra peritoneal (i.p.) injection of the opioid analgesic buprenorphine at 0.1 mg/kg (Temgesic, Eumedica Pharmaceuticals AG). After 30 minutes, they received a lethal i.p. injection of pentobarbital at 150 mg/kg (Esconarcon, Streuli Pharma SA). Mice were perfused transcardially with heparin and then with formalin 10%. After perfusion, embryos were collected and fixed overnight in formalin 10% at 4°C. For newborns up to P6, pups were decapitated and heads were fixed in formalin 10% overnight at 4°C. P8 mice were euthanized as described for pregnant mothers, before perfusion with heparin and formalin 10%, followed by immersion in formalin 10% overnight at 4°C. Heads were immersed in PBS 15% sucrose for 8 hours and PBS 30% sucrose overnight at 4°C, embedded in OCT and frozen in an isopentane bath in liquid nitrogen. Heads were cut on a cryostat (Leica DM 1850) in $20 \mu\text{m}$ sections and stored at -80°C until analysis. Before analysis, slides were stained with DAPI (1 $\mu\text{g/ml}$ in PBS) and mounted with DABCO (25 mg/ml) in a mix of 90% glycerol 10% water.

Young and adult mice (P10-P200)

Mice were euthanized as described for pregnant mothers. After perfusion, brains were collected and fixed overnight in formalin 10% at 4°C. Brains were embedded in 3% low melt agarose (TopVision Low Melting Point Agarose, Thermo Fisher Scientific®) and cut in $50 \mu\text{m}$ sections using a vibratome (Microm HM 650 V). Sections stained with DAPI, placed on Superfrost Plus slides, and mounted with DABCO.

Imaging

Slides were imaged on an epifluorescence Leica DM5500B microscope with a HC PL FLUOTAR 10x/0.30 DRY objective and a Leica Leica-K5 camera. Image acquisition and merging of tile scans was done using the LAS X software (version 3.7.4.23463). Tile scans of entire coronal sections were acquired for UV (470 nm), Cy3 (605 nm) and GFP (527 nm) channels.

Image analysis

In embryos and young mice up to P8, the expression of tdTomato was assessed visually and manually. Once multiple cells could be clearly identified within a structure, on several consecutive sections and bilaterally, expression onset was noted. After P8, relative expression levels of tdTomato were quantified by counting the proportion of DAPI-positive cells expressing tdTomato within specific brain areas. Image analysis was done using QuPath (version 0.4.3). Cells were segmented passed on DAPI staining. Next, tdTomato-expressing cells were identified using the Subcellular detection tool, set to detect both red (tdTomato) and green (autofluorescence) signals. All cells with autofluorescence signals were excluded from the analysis.

IHC on free floating sections

Mice were anesthetized by an i.p. injection of pentobarbital (200 mg/kg) and perfused transcardially with heparin and then with formalin 10%. Brains were kept overnight in the same fixative at 4°C and then embedded in agarose low melt (Thermo Fisher Scientific®) to be cut in 50 µm sections using a vibratome. Slices were post-fixed 10 minutes in formalin 10% at room temperature and washed in 1x PBS. Sections were pre-incubated in incubation solution (1x PBS, Triton X-100 0.5%, FCS 10%) for 1 hour at room temperature with agitation. Primary and secondary antibodies were diluted in incubation solution. Sections were incubated with primary antibody overnight at 4°C with agitation. Sections were washed 3×15 in washing solution (1x PBS Triton X-100 0.5%) before incubating for 3 hours at room temperature with a secondary antibody. Sections were washed 3×15 in washing solution, and nuclei were stained with DAPI (1 µg/ml in PBS) for 5 minutes at room temperature. Slides were mounted with DABCO (25 mg/ml) in a mix of 90% glycerol and 10% water. The double IHC was performed similarly but mixing both primary and secondary antibodies. When only endogenous fluorescence was observed, mice were prepared as for IHC on free-floating sections, but without any incubations except DAPI.

IHC antibodies

IHC imaging (Figure 6 [↗](#))

Fluorescence microscopy

Tile scans of coronal sections were made using a Leica DM5500 microscope. Stitching was performed with the build-in Tilesan module of the LAS X software. For expression screening experiments, specific tdT fluorescence was controlled by verifying the absence of autofluorescence in the green channel.

Confocal microscopy

Confocal images were acquired using a Leica SP8 confocal microscope and deconvolved with the build-in Hyvolution module.

Viral injections (Figure 6 [↗](#))

Injection

A 4-month-old Tg(Smim32-Cre)61Irod female mouse was anesthetized with isoflurane (3–5% induction, 1–2% maintenance), and the skin overlaying the skull was removed under local anesthesia. Mice were then head-fixed with ear bars and a nose clamp on a stereotaxic apparatus (Stoelting). Eyes were protected from drying with artificial tears. The body temperature was monitored with a rectal probe and was maintained at ~37°C using a heating pad (FHC) during surgery. Bilateral craniotomies were performed above the CLA using an air-pressurized drill. Glass pipettes filled with solutions containing AAV2-Ef1a-DIO-hChR2-(E123/T159C)-EYFP were inserted in one hemisphere at a time. Antero-posterior (AP): 1.1 mm, medio-lateral (ML): $\pm$ 2.85 mm relative to bregma and dorso-ventral (DV): 2.4 mm relative to brain surface. A volume of ~100 nL was slowly injected. The pipette was left in place for a period of 3–5 min before its removal and the suturing of the skin. All injections were bilateral unless specified.

Analysis of viral expression

For evaluation of viral expression in the CLA, the mouse was anesthetized ~3 weeks later by an intraperitoneal (i.p.) injection of pentobarbital (200 mg/kg) and perfused transcardially with 20 ml of phosphate-buffered saline (PBS, 0.1 M, pH 7.3) followed by 50 ml of 4% paraformaldehyde (PFA) in 0.1 M PBS at 4 °C. The brain was removed and left overnight in 4% PFA. After embedding the brains in 4% agarose, 40 μ m coronal slices were cut with a vibratome and collected in PBS (0.1 M). For immunostaining, slices were rinsed in PBS 0.1% TX-100, 0.7% H₂O₂. Slices were then incubated with PBS 1% BSA for 1 hour at 21–25 °C, and then with the primary antibody rabbit anti-YFP (1:1000, Invitrogen, ref. A6455), overnight at 4 °C. The day after, slices were rinsed and incubated with biotinylated goat anti-rabbit IgG (1:200, Vector, ref. BA-1000) for 1 hour at room temperature. Slices were then processed using an avidin–biotin–peroxidase complex (ABC kit, Vector Laboratories) and reacted with the chromogen 3,3'-diaminobenzidine (DAB, Sigma-Aldrich). Slices were mounted with Aquatex mounting medium.

Statistics

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Competing interests

The authors declare that they have no competing interests.

Funding

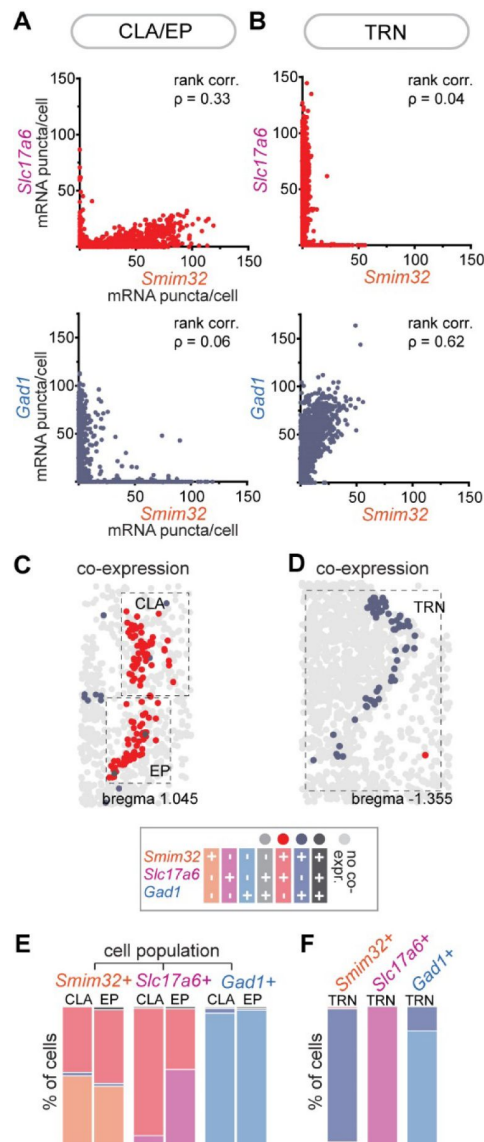
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Author contributions

Ivan Rodriguez and Alan Carleton, conceptualization, funding acquisition, supervision, writing – original draft, review & editing; Joël Tuberosa and Madlaina Boillat, investigation, project administration, formal analysis, visualization, writing – original draft; Julien Dal Col, methodology; Leonardo Marconi, investigation, formal analysis; Julien Codourey, Loris Mannino, Elena Georgiou, Marc Menoud, investigation.

Supplementary figures



Supplementary Figure 1

Smim32 is expressed in excitatory neurons in the CLA and in inhibitory neurons in the TRN.

(A,B) Correlation between expression levels of *Smim32* and *Slc17a6*, and *Smim32* and *Gad1*, in cells from the (A) CLA/EP and from the (B) TRN. Each dot corresponds to one cell. All cells from both hemispheres, within the regions of interest (ROI, dotted line in (C,D)), were included in the analysis. p value indicates Spearman's rank correlation. CLA/EP, $n = 6272$ cells; TRN, $n = 11101$ cells.

(C,D) Illustrative map of cells co-expressing *Smim32* with *Slc17a6* (red) or *Smim32* with *Pvalb* (purple grey) in the CLA/EP and in the TRN. A cell was considered positive for a marker when a minimum of 20 mRNA puncta were attributed to this cell. The dotted lines highlight the ROIs used for quantification analyses shown in (E,F).

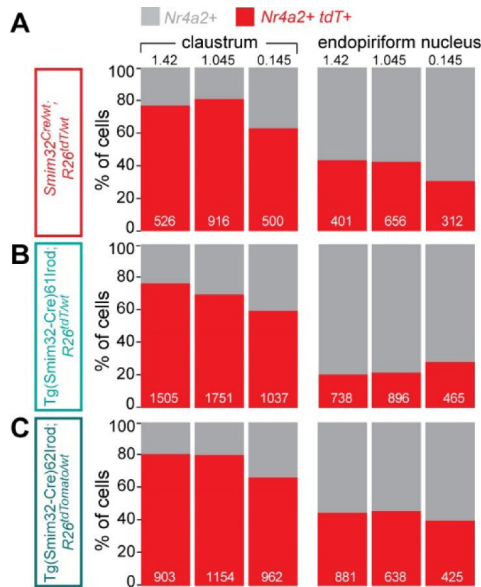
(E,F) Bar plots showing the percentages of cells positive for one or several markers. The analysis was done separately for each cell population.

Supplementary Figure 2

Proportion of CLA neuronal population expressing Cre driven reporter.

(A) Quantification of *tdT* expressing cells in *Nr4a2*-positive neuronal population in *Smim32^{Cre/wt}; R26^{tdT/wt}* mice. Bar plots show the percentage of *Nr4a2*-positive cells from the CLA and EP that co-express *tdT*. Numbers in the bar plots indicate the number of cells analyzed in the corresponding region and anteroposterior position. n= 1 *Smim32^{Cre/wt}; R26^{tdT/wt}* mouse.

(B-C) Same as above, but for *Tg(Smim32-Cre)61Irod* and *Tg(Smim32-Cre)62Irod* mice. n= 2 *Tg(Smim32-Cre)61Irod* mice, n=1 *Tg(Smim32-Cre)62Irod* mouse.



Supplementary Figure 3

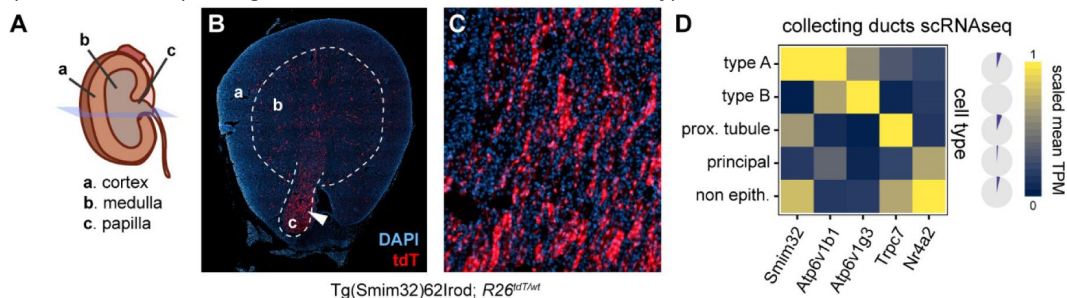
Smim32 expression in the kidney

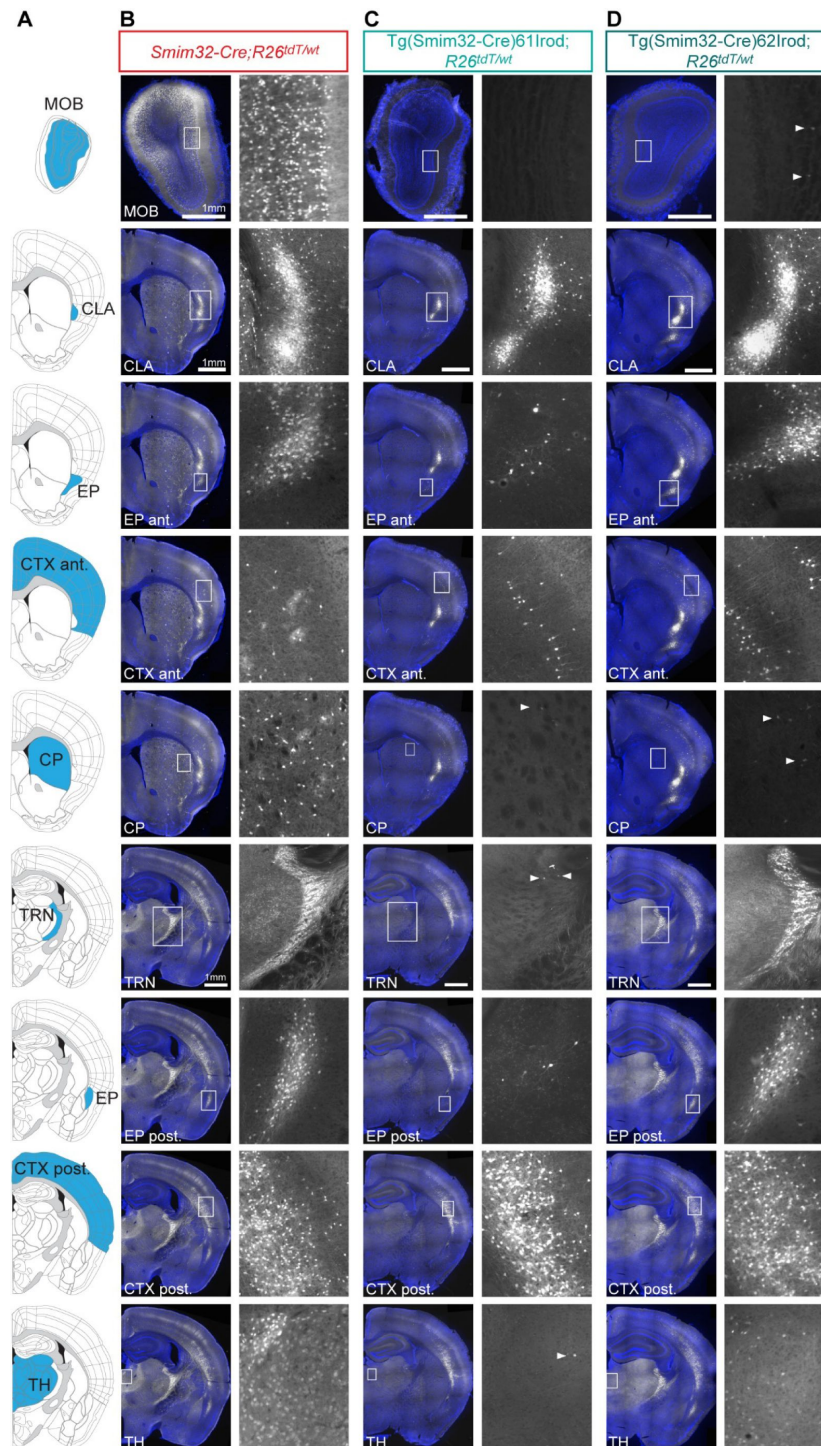
(A) Schematic of a mouse kidney with the cutting plane of the section shown in (B) and (C).

(B) Reporter expression in the kidney of a *Tg62(Smim32-Cre)* mouse.

(C) Zoom on the papilla, where arrays of reporter-expressing cells can be observed, likely forming the walls of distal tubules.

(D) Re-analysis of mouse collecting duct scRNAseq data from Chen et al.⁴⁰. Statistics were computed for cell types (rows of the matrix) and genes of interest (columns of the matrix). For each gene, mean expression values (mean TPM) were divided by the maximum observed mean expression across all samples to produce the scaled mean TPM value (max mean TPMs: *Smim32*=0.216, *Atp6v1b1*=353, *Atp6v1g3*=3088, *Trpc7*=2.06, *Nr4a2*=8.35). On the right, the dark areas of the pie charts represent the proportion of cells expressing more than 1 TPM of *Smim32* for each cell type.





Supplementary Figure 4

Representative images of Cre-driven tdT expression in *Smim32* transgenic mice

(A) Schematics of the regions of interest within which tdT expression was quantified.

(B-D) Representative epifluorescence images of tdT expression in *Smim32-Cre*, *Tg(Smim32-Cre)61Irod* and *Tg(Smim32-Cre)62Irod* mice carrying the Cre-dependent *R26-tdT* allele. Left panels, overview of the section stained with DAPI (blue) and endogenous tdT fluorescence (white). Scale bar: 1 mm. The white square highlights the region magnified in the right panels. White arrowheads point towards tdT expressing cells in regions where expression is very sparse.

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Reviewer #1 (Public Review):

Summary:

The manuscript by Tuberosa et al outlines the generation of a new set of transgenic mice that express different recombinases specifically in Smim32 positive cells. They show that Smim32

is a useful marker of the mouse claustrum. Therefore, these mice could be useful for functional studies focused on measuring claustrum activity or manipulating the claustrum using optogenetic and pharmacogenetic tools.

Strengths:

The manuscript provides a new genetic approach to target claustrum neurons, using Smim32. The work may help future studies where claustrum excitatory neurons are measured or manipulated.

Weaknesses:

A toolbox is only useful if others can use it. Therefore, these mice should be made available to the community through commercial vendors. Without this added step, this toolbox and method does not provide any utility to the research community.

The data presented and quantified in each figure subpanel are from N = 1 mouse. This is not acceptable or conventional. Replication is an important aspect of any paper, and currently, there are no replicates contained in the manuscript. Additional examples of female mice should also be included and separately quantified. Mice from different litters should be used for replicates.

Given the preliminary nature of these data from the minimum possible number of mice, a better characterization of all data should be undertaken.

The tone of the paper implies that this is the superior way to locate the claustrum. A more balanced discussion of the strengths and weaknesses of these mice should be included. Several sentences highlighting the shortfalls of other approaches are overstated and should be toned down.

<https://doi.org/10.7554/eLife.99168.1.sa3>

Reviewer #2 (Public Review):

Rodent studies of claustrum are complicated by the tube-like shape of this nucleus. As such, judicious viral strategies alone or in combination with existing Cre driver lines (Egr2, Gnb4, Slc17a6, and Tbx21) represent the current gold standard for claustrum structure and function investigation. Any improvement in tools that would allow better genetic access to the claustrum are always desired, as with any nucleus in brain. This paper describes the expression pattern of the gene Smim32 and characterization of new mouse transgenic lines expressing Cre/Flp recombinase driven by the Smim32 promoter. The authors should be applauded for the work to develop these new tools presented in the study. Overall, the strengths of the paper lie in the development of new mouse lines that are well-characterized in comparison to other molecular markers of the claustrum. Weaknesses lie in poor anatomical definitions of the claustrum (and endopiriform nucleus). Smim32 expression is used to define claustrum anatomical boundaries, rather than first using several structural, molecular, and connectivity lines of evidence to define the claustrum anatomically and then to assess whether Smim32 expression fits within this anatomical definition. Another major weakness is the fact that Cre/Flp expression driven by the Smim32 promoter is present in non-claustrum regions, including the neighboring cortex, striatum, and endopiriform nucleus as well as the more distant thalamic reticular nucleus. Despite this, the conclusion of the study, as communicated by the authors, is that selective interrogation of the claustrum is now possible with these Smim32-based tools. Therefore, the data do not support the claims and conclusions.

Very concerning is problematic language in the abstract and introduction sections that diminish the impact of several published studies (not cited) that have led to important findings regarding claustrum function. The authors Create an argument that all the research performed thus far on the claustrum is unreliable because targeting the structure has been sub-optimal. This is definitely not the case for several studies from multiple labs. If investigators new to the claustrum were to read this paper, they would conclude that all previous data hold little-to-no value and that using these tools set forth the possibility, at long last, to solve claustrum structural and functional queries. Here is an example from the abstract of the problematic language: "However, research on the CLA has been challenging due to difficulties in specifically and comprehensively targeting its neuronal populations. In various cases, this limitation has led to inconsistent findings and a lack of reliable data." (no references cited). Since Smim32 driven recombinase (in 61 or 62lrod) is not exclusively expressed in the claustrum, it is not clear how Smim32 is an advantage over possible Nr4a2 or, the more selective, GNB4 Cre driver lines. Taken together, the goal of the study as articulated in the Introduction: "Our goal was here to generate genetic tools capable of targeting the majority of mouse CLA projection neurons without affecting other brain cell populations, or tissues outside the brain" has not been met and, therefore, the conclusion of the study based on the data "With these genetic tools in hand, the comprehensive targeting and functional probing of the densely connected CLA is now possible" is unfortunately also unmet.

The manuscript does convincingly show that Smim32 targets excitatory neurons in the claustrum as evidenced by exclusive overlap of Smim32 expression with Vglut2 and not GAD (fig 1 and suppl fig 1). Additionally, the manuscript provides sufficient evidence that neurons in the claustrum area expressing Smim32 further co-express a number of other molecular markers of claustrum, including Nr4a2 (fig1), Lxn, Gnb4, and Oprk1 (fig 2), and Slc17a6 (suppl fig 1). The authors further show that Smim32 is not co-expressed with molecular markers of layer VI cortex like Ctgf and Rprm (fig 2). However, by limiting the line of evidence to molecular expression, the study fails to escape the limitations of molecular markers, which cannot by themselves be used to define the anatomical boundary of the claustrum. The expression of several of these markers in the neighboring endopiriform nucleus, including Smim32, is evidence that using molecular markers as a sole indicator of the anatomy of the claustrum is not warranted.

While the anatomical boundaries of the claustrum remain somewhat debated, several standards have emerged to delineate claustrum boundaries. These include immunoreactivity against Gng2 (or PV, especially in rat) to indicate claustrum or against Crym to counter-indicate claustrum. In addition, injection of retrograde tracers into the anterior cingulate cortex or retrosplenial cortex, for example, results in selective targeting of (large) subpopulations of claustrum neurons that help define claustrum location. Further targeting of neurons projecting to the anterior insula or thalamus has been used to delineate the boundaries of what some consider the claustrum shell and others consider the deep layers of the insula. The use of any of these approaches to delineate the claustrum anatomy should be used to describe the spatial distribution of Smim32 and Cre or FlpO in the transgenic lines.

The manuscript provides a description of Smim32 promoter-driven tdTomato in the three transgenic Cre lines during development. This shows strong expression in claustrum and not in surrounding regions. However, as the claustrum borders are not distinct without markers, the anatomical boundary of claustrum for this analysis is deemed arbitrary - an issue that is exacerbated when looking at the developing brain where atlases are less precise and boundaries of the claustrum are ill-defined.

<https://doi.org/10.7554/eLife.99168.1.sa2>

Reviewer #3 (Public Review):**Summary:**

In the manuscript by Tuberosa et al., the authors set out to identify a genetic marker for the claustrum to create transgenic mice as tools to study this challenging brain region. To achieve this, the authors first re-analyzed published scRNAseq datasets from mouse frontal cortex and identified a unique cluster expressing Smim32, which correlated with Nr4a2, a previously reported claustrum marker (though also expressed in layer 6 and elsewhere). Importantly, Smim32 was also found to strongly express in the layer 6 and the thalamic reticular nucleus (with weaker expression in other parts of cortex, striatum, thalamus, olfactory bulb and more). The authors then extensively characterize Smim32 expression relative to a few other genes associated with claustrum and layer 6, as well as creating several novel transgenic mice focused on the Smim32 gene.

Strengths:

The main strength of the paper is the well done scRNAseq analysis, the beautiful ISH images/reconstructions, and the assessment of gene expression throughout development. The main value of this paper is adding the Smim32 gene to the list of markers expressed in the claustrum, though it is not specific to the claustrum, showing extensive expression in TRN and layer 6 of cortex.

Weaknesses:

The main weaknesses are that the results do not support the conclusion, namely that the Smim32 gene is not specific to the claustrum and that no other orthogonal approaches were used to define the claustrum, such as retrograde neuroanatomical tracing from cortex. Also, these results are of limited applicability as the gene expression was only performed in mice, so it is unclear how Smim32 relates to claustrum in other mammalian species (e.g. primates), which have a very clearly defined claustrum. The article is also missing some key literature on the anatomical definition of claustrum, specifically as it relates to the endopiriform nucleus (which is putatively considered part of the claustrum in rodents).

<https://doi.org/10.7554/eLife.99168.1.sa1>

Author response:**Data replicability**

| *There are no replicates contained in the manuscript. (Reviewer #1)*

We respectfully disagree with this statement. In this manuscript, we included both cell and animal replicates. For cell replicates, we analyzed over 50,000 cells using RNAscope and over 10,000 cells using RNAseq, employing two independent methods on diHerent animals. We believe this extensive analysis is sufficient by any standards. Regarding animal replicates, we generated four diHerent transgenic lines (two knockin lines and two BAC transgenic lines), which is an uncommon and rigorous effort. We analyzed dozens of animals, consistently observing the expression pattern of Smim32 and its derived transgenes across multiple experiments, including crosses between transgenics and various reporter lines, which is again an uncommon and rigorous effort. These experiments were conducted on animals from diHerent litters to ensure robustness. Additionally, our longitudinal study, which includes 13 animals harvested at two-day intervals from E16 to P20, provides further consistency of our data.

However, to underscore the consistency of endogenous Smim32 expression, when submitting a revised manuscript, we will present Smim32 expression levels across individuals in single-cell RNA-seq data. Furthermore, we will pool data from diHerent transgenic animals to demonstrate interindividual variability in the claustrum of adult animals.

Additional examples of female mice should also be included and separately quantified.
(Reviewer #1)

We initially analyzed both males and females for one line (the Smim32-Cre knock-in line). Since we observed no diHerences between males and females (which we will note in the revised manuscript), we subsequently limited our analyses to males to minimize the use of animals.

Claustrum definition

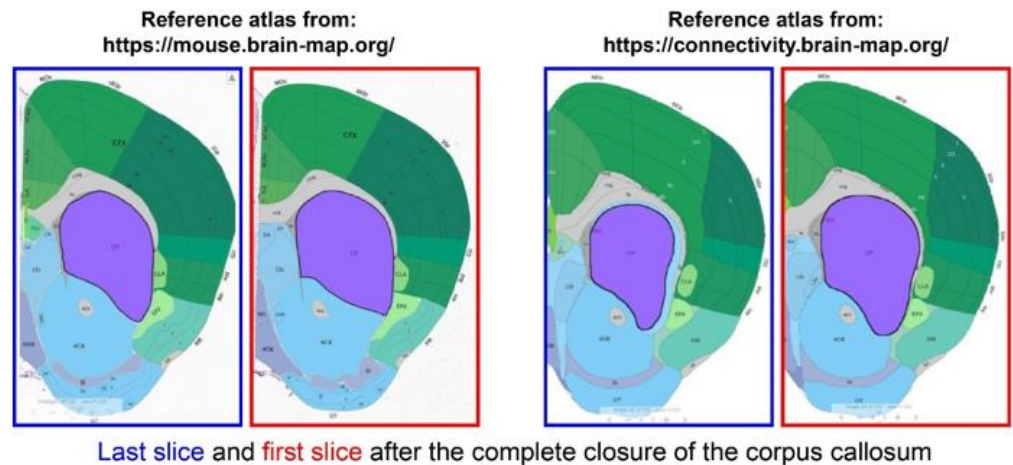
Weaknesses lie in poor anatomical definitions of the claustrum (and endopiriform nucleus). (Reviewer #2)

No other orthogonal approaches were used to define the claustrum, such as retrograde neuroanatomical tracing from cortex. (Reviewer #3)

We share the reviewers' opinion that the claustrum (CLA) and endopiriform nucleus (EN) are poorly defined anatomically in rodent brains due to the limited development of white matter tracts. This ambiguity has led to many conflicting descriptions of CLA/EN boundaries in various papers and atlases, including those by Paxinos and the Allen Brain Institute. Notably, the Allen Institute frequently updates the shape and anatomical location of the CLA/EN in their reference atlas, resulting in diHerent websites displaying various versions (as illustrated in rebuttal figure 1 at comparable levels of the anteroposterior brain axis). It remains uncertain which version would most eHectively satisfy the entire scientific community, if any. Indeed, after many years of working on these structures and surveying the literature, we regret to note that there is currently no consensus on the anatomical definition of the CLA and EN, even among expert laboratories using tracing or staining methods. At one end of the spectrum, some authors define the CLA as a small nucleus that could be, for example, characterized by the PVrich plexus. At the other end, other authors consider it part of a larger complex that includes the EN and extends dorsally to the S2 cortex. Additionally, diHering definitions of the core and shell regions, as well as the precise anteroposterior extent of the nucleus, further complicate the issue.

Author response image 1.

Comparison of CLA and EN shapes in two recent versions of the Allen brain atlas



Given this lack of consensus, we deliberately opted for a molecular definition of the claustrum and its projection neurons. We used a set of well-documented canonical markers for the claustrum and neighboring neurons to determine the expression pattern of *Smim32*. The claustrum-specific markers we selected (*Nr4a2*, *Lxn*, *Gnb4*, *Car3*, etc.) have been extensively studied and allow us to distinguish claustrum projection neurons from neighboring and intermingled populations. Although none of these individual markers are exclusively specific to CLA and EN neurons, the combined expression of these markers provides greater confidence in identifying the diHerent neuronal populations in space.

Smim32 expression is used to define claustrum anatomical boundaries, rather than first using several structural, molecular, and connectivity lines of evidence to define the claustrum anatomically and then to assess whether Smim32 expression fits within this anatomical definition. (Reviewer #2)

Contrary to the reviewer's suggestion, we do not define the claustrum based on *Smim32* expression. Instead, Figures 1 and 2 demonstrate that *Smim32* expression is highly correlated with the expression of known claustrum markers (*Nr4a2*, *Lxn*, *Gnb4*, *Car3*, etc.), both regionally and at the cellular level. As suggested by Peng et al. (2021, Fig. 4 and Extended Data Fig. 11), this population of cells, which includes the claustrum, a specific subset of cells in cortical layer 6, and the dorsal endopiriform nucleus, forms a discrete group of neurons sharing the same transcriptomic identity. Given what is known about the connectivity of claustrum and endopiriform nucleus projection neurons, this population obviously includes neurons projecting to various areas, likely fulfilling distinct functions. Whether these cells should be subdivided based on projection area, developmental origin, or structural features is beyond the scope of this article.

Specificity issues

Cre/Flp expression driven by the Smim32 promoter is present in non-claustrum regions, including the neighboring cortex, striatum, and endopiriform nucleus as well as the more distant thalamic reticular nucleus. (Reviewer #2)

The Smim32 gene is not specific to the claustrum. (Reviewer #3)

We do not claim that endogenous *Smim32* expression is exclusive to the claustrum or that the knock-in lines, by themselves, are sufficient to isolate claustrum neurons without combined approaches based on the transgenic lines presented here. However, there are significant diHerences in the expression pattern between endogenous *Smim32* and the expression of Cre

in the various derived transgenic lines, which might not have been clear in the current manuscript. Notably, there is no expression of Cre in the striatum and the thalamic reticular nucleus, and only sparse expression in the endopiriform nucleus in Tg61(Smim32-cre). Each transgenic line provides different levels of overlap with the endogenous Smim32 expression, with the Tg61(Smim32-cre) line allowing for the most specific genetic access to claustrum neurons. Again, for greater specificity, any of these lines could be used in combined approaches, such as viral targeting (as shown in Figure 6A and B) or using transgenic intersectional (dual recombinase) approaches based on Cre- and Flp-expressing mice with an overlap in the claustrum, leading to circuit-specific and/or claustrum-only labeling.

This means that our claims are supported by the observed data. However, we acknowledge that we may not have clearly explained the specificity of the random transgenes, which could have led some reviewers to believe that « the data do not support the claims ».

We will clarify these points in the revised manuscript and include additional examples and quantifications to highlight the differences between endogenous Smim32 expression and Cre expression in the transgenic Tg61(Smim32-cre) line.

Regarding Cre-expressing cells in the neighboring cortex (layer 6 projection neurons), these cells are genetically distinct from other layer 6 cortical neurons and express the same canonical markers as claustrum projection neurons, likely sharing also the same transcriptomic identity. We will provide a more detailed characterization of these cells in the revised manuscript.

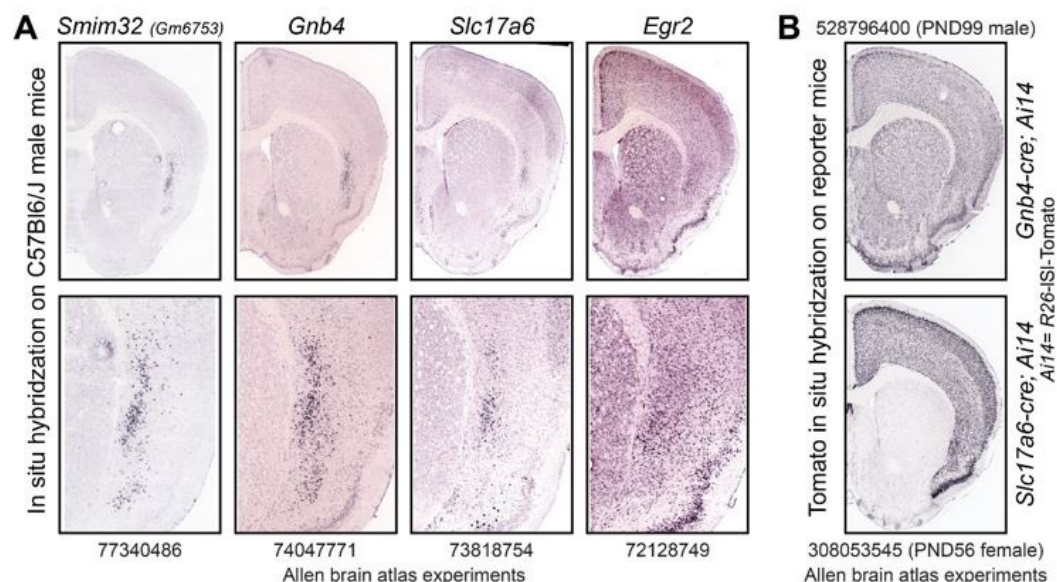
Since Smim32 driven recombinase (in 61 or 62lrod) is not exclusively expressed in the claustrum, it is not clear how Smim32 is an advantage over possible Nr4a2 or, the more selective, GNB4 Cre driver lines. (Reviewer #2)

Over the years, we have found a limited number of Cre lines used in the literature for targeting claustrum neurons. These include Gnb4-cre, Slc17a6-cre (also known as Vglut2-cre), Egr2-cre, Tg(Tbx21-cre), Ntng2-cre, Cux2-cre and Esr2-cre lines. We have not found any study describing and/or using an Nr4a2-cre line. Although a Nr4a2-Dre line exists (that we have studied in our laboratories), caution is warranted in its use, as it lacks the complete coding sequence of the Nr4a2 gene.

One problem with Nr4a2 is its documented expression in the adjacent Layer 6b cortical neurons, which discards it as a suitable candidate to selectively target the claustrum. Furthermore, Nr4a2 is also expressed in a majority of the endopiriform nucleus neurons, whereas endogenous Smim32 is expressed in a smaller proportion of these cells, and is restricted mainly to the dorsal endopiriform nucleus. These reasons led us to select Smim32 over Nr4a2.

Author response image 2.

(A) In situ hybridization for various CLA/EN marker genes. (B) Developmental recombination observed outside the CLA/EN in various cre lines (all data from the Allen brain databases)



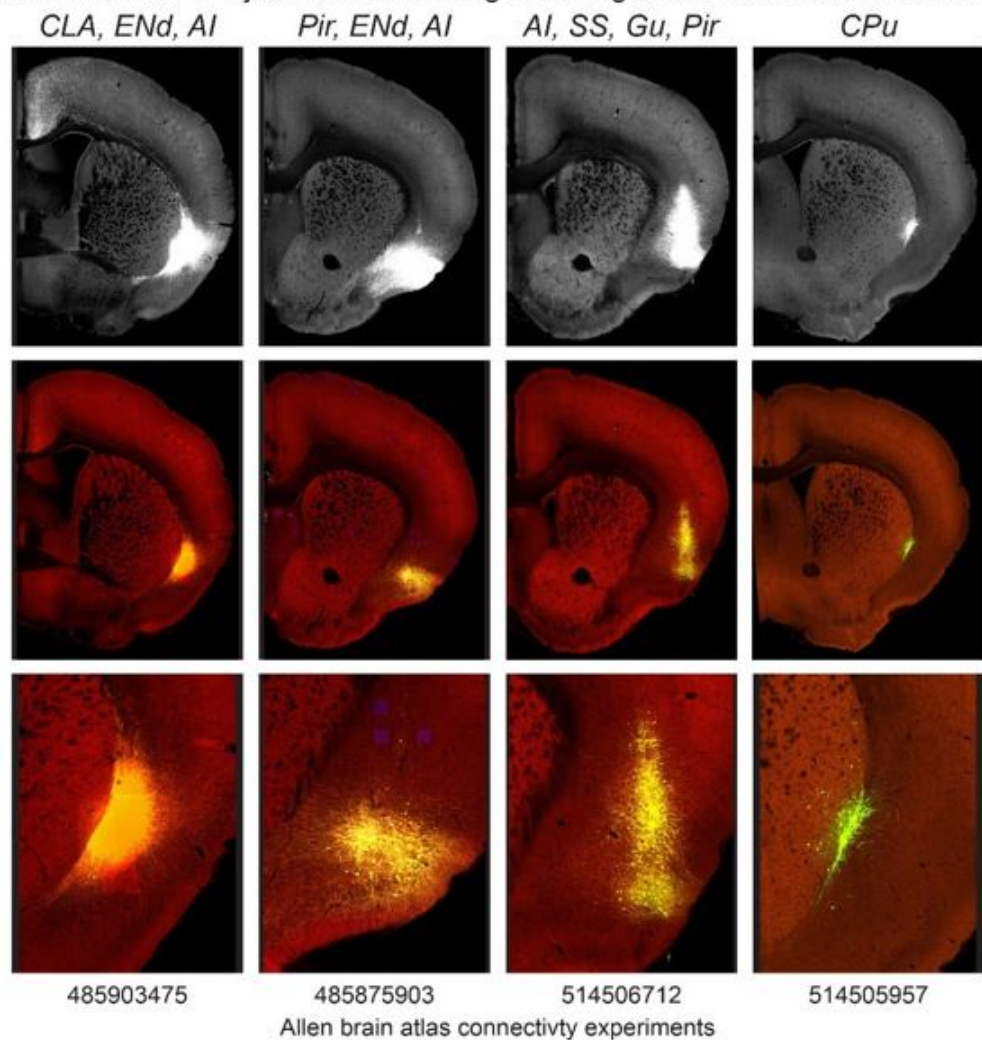
What are the advantages of using the different *Smim32*-cre lines over the existing *Cre* lines mentioned above?

Let's first consider the *Gnb4*-cre line, which is considered one of the best available. Although the endogenous *Gnb4* gene appears to have a similar expression pattern to *Nr4a2*, *Slc17a6*, and *Smim32* in the striato-claustro-insular region of adult mice (Rebuttal Figure 2A), the results observed with the *Gnb4*-cre line either shows otherwise, or indicate that the *Cre* line does not fully recapitulate *Gnb4* endogenous expression (Rebuttal Figure 3). Indeed some neurons in the insular cortex, piriform cortex, and putamen express the *Cre* recombinase (possibly due to low *Gnb4* expression not detected in the in situ hybridization data of the Allen brain institute or due to nonspecific transgene expression) and will recombine viral vectors injected in adult mice (Rebuttal Figure 3). Therefore, this *Cre* expression outside the CLA/EN neurons in the *Gnb4*-cre line presents complications for data interpretation, depending on the viral injection coordinates and the quantity of injected vectors.

Author response image 3.

Specificity of the *Gnb4*-Cre line tested with viral transduction in adult mice (all data from the Allen Brain Institute database). The top and middle rows display the same data but with different scaling of the lookup tables to highlight either the patterns of axonal projections (top) or the infected neurons themselves (middle). The bottom row shows a higher magnification of the infection site. Note that individual neurons cannot be resolved in experiment 485903475 due to signal saturation.

AAV-DIO-EGFP injected the following brain regions of adult *Gnb4-cre* mice



Cre expression in the CLA appears more specific in the various *Smim32-cre* transgenic lines than in many of the lines mentioned above. Although we have no doubt that the different existing transgenic lines can target CLA neurons, the selectivity of the targeting (for example, the fraction and types of CLA neurons versus potential non-CLA neurons) remains to be fully described for most of the lines. It is particularly true in the case of *Tbx21* and *Esr2* (used as drivers for the *Tg(Tbx21-cre)* and *Esr2-cre* transgenic lines). *Tbx21* is not endogenously expressed in adult CLA neurons (evaluated by in situ and RNAseq data) and *Egr2*, if expressed in the claustrum, is not restricted to CLA neurons as it is an immediate early gene expressed in recently active neurons (Rebuttal Figure 2A).

Cre expression in the EN is observed in all Cre-expressing transgenic lines used to target the claustrum (with the exception of *Slc17a6-cre*). This can naturally be problematic for some approaches. Luckily, the random integrant *Tg61(Smim32-cre)* we describe in our manuscript shows a strong expression in the claustrum, and very limited expression outside the CLA (a very weak activity in the EN), representing a novel tool with improved claustrum selectivity. An advantage of the *Tg61(Smim32-cre)* over the *Slc17a6-cre* is that more CLA neurons can be targeted with the *Tg61(Smim32-cre)* line.

Another advantage of our four transgenic lines is their versatility; they can be used to recombine reporter lines as well as FRT-floxed and loxP-floxed knockouts in limited neuronal

populations. They will be employed in the future for intersectional genetics to exclusively target CLA neurons. Existing transgenic lines cannot offer these possibilities because their marker genes are broadly expressed in the brain during embryogenesis, leading to the impact on a large number of non-CLA/EN neurons. This is evident in the Gnb4-cre and Slc17a6-cre lines crossed with the Ai14 reporter line expressing the fluorescent protein tomato (Rebuttal Figure 2B, right panels). Similar observations have been made for the Ntn2-Cre and Cux2-cre lines (see the Allen Brain Institute database for these data). Alternatively, inducible recombinase systems, such as the Gnb4-IRES2CreERT2-D line, could be used. However, the Gnb4-IRES2-CreERT2-D line requires tamoxifen to induce Cre recombination, which can be problematic depending on the research context, as well as recombinations in the absence of tamoxifen treatment (see experiments 560948627 and 560948194 in the Allen Brain database).

It is unclear how Smim32 relates to claustrum in other mammalian species (e.g. primates) (Reviewer #3)

As mentioned in the last paragraph of the introduction of the initial manuscript, Smim32 is specifically expressed in the claustrum of a primate species, Homo sapiens (reference 37 of the initially submitted manuscript).

Availability of the transgenic mice

These mice should be made available to the community through commercial vendors. (Reviewer #1 and #2 in private comments)

We are pleased to see that two of the three reviewers would like to see these mice available. These mice will not be kept for ourselves, and we will distribute them at some point in time, but this will naturally occur after the publication of the revised manuscript.

Critical comments on discussion and other topics

A clear description of the search in the Allen Mouse Brain Atlas is missing. A search for Smim32 in the ISH mouse atlas did not provide any hits and so it would be useful to include in the methods or results section the exact query used for examination of Smim32 expression as well as other genes identified in this process. (Reviewer #2)

Smim32 has been referred to by different names in various versions of the mouse genome. For the readers not versed in navigating genomes and annotations, before being officially named Smim32, this gene was originally called Gm6753 (as noted in the Allen Brain Institute database, see Rebuttal Figure 2A for an example of their in situ data) and later Gm45623.

Several sentences highlighting the shortfalls of other approaches are overstated and should be toned down. (Reviewer #1)

Very concerning is problematic language in the abstract and introduction sections that diminish the impact of several published studies (not cited) that have led to important findings regarding claustrum function. The authors create an argument that all the research performed thus far on the claustrum is unreliable because targeting the structure has been sub-optimal. (Reviewer #2)

A more balanced discussion of the strengths and weaknesses of these mice should be included. (Reviewer #1)

We regret if our choice of language inadvertently appeared to undermine the contributions of our colleagues; that was certainly not our intention. The paragraph in question was meant

to address certain studies that we believe have led to inconsistent findings and unreliable data due to a lack of rigorous methodology in targeting claustrum projection neurons. To avoid singling out specific works, we chose not to cite them directly. We understand that some colleagues whose research does not fall under the “various cases” mentioned may feel unfairly targeted by this statement. We will revise this section to better clarify our intent and ensure it is respectful of all contributions. We will rephrase passages in the abstract, introduction, and discussion to provide a balanced view of the strengths and weaknesses of these mice.

Our main goal is to provide tools to specifically target claustrum cells based on their transcriptomic identity, which we believe is the best means to assess the function of any neuronal population. Due to the intermingling of claustrum neurons with neighboring populations, employing stereotaxic injections in the claustrum without genetic segregation will always infect and label physically adjacent cells that do not belong to the claustrum, ontologically and functionally speaking.

Similarly, targeting claustrum neurons retrogradely by injecting into claustrum projection sites likely labels neurons from diHerent populations. For instance, as reviewer 1 mentions Erwin et al. (2021), infecting retrosplenial projections without genetic specificity labels many claustrum Synpr+ neurons (considered the claustrum core), a small proportion of claustrum Nnat+ neurons (considered the claustrum shell by some, and non-claustrum neurons by others), and some neighboring cortical L6b neurons. These three populations have very diHerent transcriptomic identities, connectivity patterns, and likely distinct functions.

Thus, we believe that genetic specificity provides an important added value for selectively targeting the claustrum or claustrum-insular complex.

| *A better characterization of all data should be undertaken. (Reviewer #1)*

Having generated hundreds of transgenic lines over the years, we have never performed a more thorough analysis of transgenic lines, nor have a recollection of reading a publication evaluating at such a precise level the expression pattern of transgenes in mice. We, therefore, do not see exactly what the reviewer means by this remark. It is possible, not being native English speakers, that we did not grasp a certain form of joke.

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